INFORMATION SYSTEMS 7315 WORK INTEGRATED LEARNING

TASK 1

The SOCIETY FOR ANIMALS IN DISTRESS





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GitHub Link:
https://github.com/EMGPMD/insy7315-2026-task-1-st10281928-ndivhuwondou

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9.1.4 Introduction

The background and operational environment of the organisation.

The Society for Animals in Distress (SAID) is a non-profit animal welfare organisation whose activities are directed towards the improvement of animal health, animal welfare and quality of life for animals under their care (The Society for Animals in Distress, 2025). Veterinary and clinical work is a crucial aspect of SAID's services, but the day to day work of the organisation is not just about clinical work, but also includes a wide variety of interconnected activities in the hospital, with animals, outreach, administration and supporting activities. The hospital environment, animal care, field and mobile outreach, reception and administration, pharmacy functions, ward work, kennel management, theatre preparation, admissions, equipment inspection, cleaning tasks, staff relief and other operational tasks all make up the operational environment of the organisation (The Society for Animals in Distress, 2025). The activities involve the synchronisation of duties between employees and supervisors across various geographical and temporal locations and the proper execution of key tasks.

Staff members cannot only be considered by working days. The organisation should also think about what each member of staff will need to do, when they will need to do it, where they will be doing it, if they will be needing another employee to help or if they will be needing another skill or role, and if the responsibility has been carried out or not.
This establishes a scheduling situation where there is a strong tie between scheduling and daily operations (RotaPlanner, 2026) . A roster is a snapshot of the desired state of operations, but keeping it updated with status updates of tasks, reassignments, notifications and supervisor intervention will keep it in line with the actual status of operations during the working day (RotaPlanner, 2026) .

As a result, the project recognises workforce coordination as a key operational need for SAID and sees a potential to enhance the current process with a dedicated digital solution. The current operations and coordination of the hospital workforce.Current hospital operations and hospital workforce coordination.
We (the project team) discovered that a key issue that impacts SAID's hospital operations is the manual generation of employee schedules. The Friday roster is supplied and is an operational representation of the complexity the organisation has to deal with and includes 107 entries for hospital and operational duties (RotaPlanner, 2026) .

The roster shows that there are scheduling needs beyond the traditional employee shift scheduling needs for SAID. Documented responsibilities include reception and administration, pharmacy, animal wards, kennel management, theatre related, field/mobile outreach, equipment checks, cleaning, patient admissions, staff tea/lunch relief, and other operational responsibilities.

There are a number of these responsibilities with varying time requirements. There are activities which happen at a set time and staff therefore need to be fit to be at a specific time during the working day. Other activities are given a weekly pattern and must be represented consistently throughout the roster periods. Some activities might involve more than one staff member, and others might need to be carried out in conjunction with other activities. Therefore, relationships between employees, tasks, location, time and operational requirements must be taken into consideration in the operational roster. As the number of duties and changes grows, this adds an increasing amount of complexity to the manual management.

It was also determined that the operational conditions may change after the initial rostering is created. An existing roster can be subject to a variety of factors that impact its accuracy, including staff shortages, changes of availability, urgent operational needs, reassignment of roles and additional work that needs to be done that day. As a result, the list of expected activities at the start of a day might not reflect the actual situation at a later time that day. The existing roster and complexity of operations are analyzed.

The provided roster is evidence of scale and operational interdependence of the scheduling problem. A Friday roster with a total of 107 scheduled entries shows that supervisors could have a lot of individual responsibilities to coordinate over a relatively short period of time (RotaPlanner, 2026).

These entries are not only in number but also in the distinction of the duties that they represent in the roster. A theatre-related responsibility may not have the same characteristics in terms of time or staffing as a cleaning task or field-outreach activity, and a reception responsibility will have other operational requirements from a kennel duty.
The roster can also have tasks which need multiple employees to be present at the same time. If more than one person has been allocated to a specific task, the loss of one employee can have an impact on the ability to complete the task. Likewise, when one worker is shifted from one job to another, another scheduling issue could arise in another part of the roster.

This gives rise to a chain of dependencies for operation, such that one change may impact many responsibilities. These effects may be identified by supervisors on a manual roster, adding to administration time and the chance of human error.
The project, therefore, recognizes the necessity of the proposed system to offer greater than just a digital facsimile of the roster. It should aid supervisors in comprehending the operational implications of change, and warn of change when there is a scheduling conflict or unfilled responsibilities.

Identified Operational Problem

The client engagement revealed that the most challenging operational issue is maintaining an up-to-date, coordinated and visible picture of SAID's workforces with manually created rosters (Weyers, 2022). In the different operational areas, there are many repetitive, time-critical, common and interdependent functions that are part of SAID's operational tasks.

The supervisors need to distribute available staff among tasks taking into account their responsibilities, operational areas, scheduling needs and availability.
Meanwhile, supervisors need to address staffing reductions, immediate workloads, reassignment needs and other same day activities.

The challenge is, the organisation is currently using a manual roster. The overall problem is that there is no single operating environment where roster information, staff assignments, progress on tasks, changes, notifications and operational risks can be managed together (Weyers, 2022). Other important operational issues raised during the client engagement were staff shortages, task automation, and the potential use of an alert mechanism. These concerns clearly indicate that the proposed system should be able to assist in the decision-making process of the operations and should not merely be a reproduction of the information available on paper or in an existing document that is maintained by hand.
The system should therefore give supervisors a wider picture of operational tasks and give staff a visual and easy-to-access overview of the tasks assigned to them.

Proposed SAID Operations Solution

The suggested solution is the SAID Operations: Hospital Workforce, Roster and Task Management System. The system will be designed to be an operational-management platform for the entire system that will enable it to connect with employees, schedules, operational areas, operational tasks, assignments, notifications, and the progress of the tasks (Kunz, 2025) .

The proposed solution will be divided into two interfaces, one for the supervisor and one for the staff, both of which will be accessible via the web, and the latter will also be mobile-ready. The interfaces will give users different levels of functionality, depending on the roles of each user.

The supervisor interface will give authorised users the ability to create, maintain, publish, modify and monitor operational rosters. Supervisors will be able to assign staff to recurring and ad hoc tasks, reassign tasks when operational circumstances change, determine possible scheduling conflicts, determine key tasks that have no staff assigned, communicate important changes in operational tasks, and track progress of tasks.

The staff interface will give staff a direct view of the responsibilities they are charged with. Employees will be able to safely see their own schedules, information about tasks assigned to them, and be notified of relevant events, all while updating the status of tasks they are assigned to. The desired outcome is a central operational view that enables supervisors to know what is scheduled, who is scheduled for the work, where the work is likely to take place, when the work is expected to be completed and whether it is completed.
The intended solution is thus not only to digitize the existing roster but also to streamline the operational process.

The purpose and intended operational outcomes are stated clearly, effectively, and succinctly.
SAID Operations system aims to enhance the visibility, coordination, communication and monitoring of the SAID daily activities. The system will be a single source for operational

roster and task information. This will make it easier for supervisors to work from an up-to-date copy of planned activities and not have to use multiple versions of manually maintained documents or use different communication channels.

Staff will also have direct access to their responsibilities via the system. Employees will have access to the latest schedule and task information into the staff interface instead of having to depend on a printed roster or supervisors' verbal information.
Reusable task templates will provide recurring operational tasks. This should minimize repetitive data entry when creating rosters and increase consistency in the representation of regularly occurring duties (Kunz, 2025) .

The system will also deliver operational warnings/notification. Scheduling conflicts and uncovered critical duties should be brought to the attention of supervisors so that corrective action can be taken. Any changes to assignments should be communicated to relevant staff members to minimise the risk of staff working from incorrect information.

There will be an additional level of operational visibility in the form of task-status functionality. A traditional roster will convey what is planned, while the proposed system will also indicate whether a task is scheduled, in progress, completed, or if it has not been completed.
The goal is therefore to give SAID greater visibility of planned operations as well as of the progress of the tasks.

Relationship with Existing Clinical Systems

One of the critical aspects of the proposed solution is how it connects with SAID's current clinical systems. Currently, SAID is using ezyVet for clinical related activities and animal records. The proposed SAID Operations platform will not be a replacement for ezyVet or an attempt to duplicate functionality related to clinical animal management.
The proposed system will therefore not be used to manage medical records of animals, diagnoses, history of treatment, prescriptions, veterinary clinical notes, etc. Its role will instead be on operational coordination related to the organisation's workforce and their activities (Ezyvet.com, 2026) .

It is important to note that these are separate systems, as this would otherwise make the project a clinical-record-management solution. ezyVet will continue to manage the necessary clinical and animal record management activities, while SAID Operations will concentrate on people, operational areas, schedules, tasks, assignments, notifications, progress and operational history.

Further integration between SAID Operations and ezyVet would need to be considered on its own merits. This integration would involve technical, security, business, privacy and operational evaluations and would necessitate specific approvals from SAID. Thus, it is not possible to implement direct ezyVet integration in Version 1 (Ezyvet.com, 2026) .

Version 1 is intentionally geared toward the operational coordination issue that was identified during the client engagement. The system will deliver the essential capabilities needed to oversee workforce scheduling, task allocation, communication, progress and monitoring of tasks and operations.

The system will include secure authentication and role based access control to enable only authorized users to access functionality based on their roles. It will keep staff information for operational scheduling, and provide for configurable operational areas to which tasks can be mapped.

The system will facilitate task template re-use for recurring tasks, daily and weekly roster generation, management, staff assignment/reassignment, publishing rosters, personal staff schedules, status management, notifications, scheduling conflicts, uncovering critical duties, supervisor dashboards, operational notes and history of rosters/tasks.

Version 1 will not feature clinical animal records, veterinary diagnosis, financial management, payroll, accounting, public adoption, lost-and-found reporting, or SOS reporting.

This "System Boundary" can be interpreted as a management of:

People → Places → Schedules → Tasks → Assignments → Notifications → Progress → Operational History

This boundary is essential so that development will continue to be focused on SAID's identified operational coordination problem.

9.1.5 Requirements

Requirements Overview

The requirements for SAID Operations have been based on the client engagement, the operational context identified by the project team, and analysis of the provided SAID staff roster. The requirements describe the capabilities and characteristics that the proposed system must have, in order to meet the needs of the organization for workforce coordination. The requirements are split into the Functional and the Non-Functional requirements. Functional requirements define the behaviour and capabilities the system should have; non-functional requirements specify the quality aspects the system should possess (including security, usability, performance, reliability, accessibility, maintainability, scalability, privacy, data integrity) (Nourhansaeed, 2024).

The requirements are designed to ensure a correlation between SAID's operational problem and the function proposed in the system. If an operational rule or requirement has not yet been formally approved by SAID, the requirement may be considered as a provisional one and should be validated with the client prior to the finalisation of the associated functionality (Nourhansaeed, 2024).

Identify the organisational and business requirements.

This system will need to be able to meet the operational needs of SAID for better coordination of staff duties between the hospital, animal-care, outreach and supporting operational areas (Awork.com, 2026).
The system should give supervisors a single centre where they can plan and keep track of the workforce schedules. This need is a result of the manual roster being in place and needing to reflect operational responsibilities as they exist at any given time.
The system should allow for the allocation of staff members, available for duty, to operational duties, taking into account scheduling information and operational requirements. The system should help supervisors determine when a staff member might have conflicting responsibilities or when a significant responsibility is not covered.

The system should provide support for on-going and occasional operations. Recurring activities should be able to be consistently instructed with task templates which can be reused, and unexpected operational needs should be able to be added to a current roster if needed. The system should also be able to handle operational communication. Any changes in schedules and assignments should be reported to the users who are impacted so that they can have up-to-date information.
The system should enable sufficient visibility for a supervisor to see if the tasks he/she has been given are being carried out. This requirement is the defining aspect of the proposed solution from a "standard" list of assignments, as it adds a relationship between the planning and execution of the assignment.

User Requirements

The suggested system will be mainly for the supervisors and staff members. Both groups share the same operational information, but have different responsibilities and information needs (Kraeling and Tania, 2019).

Supervisors need one central place to manage and schedule their staff, repeat tasks, assign individuals, post schedules, highlight scheduling issues, determine any missed critical tasks, communicate important information, track task progress, and access pertinent operational history.

The supervisor interface should therefore offer more management capabilities than the staff interface. Supervisors must have the ability to make operational decisions and be able to adapt to changes, and also have an appropriate control of information shared with staff.
Staff needs a more straightforward operating interface in which to carry out their work. The staff should be familiar with what they are expected to do, when to do it, where it should be done and what other information is needed to do the task.

Staff members need to be able to communicate task progress through the system; updating the status of assigned tasks. For tasks that can't be done, the system should offer a suitable method of letting supervisors know.

Responsibilities for these user groups vary, so the system should not offer the same functionality to all users, but rather be role-based (Kraeling and Tania, 2019).

Functional Requirements

Secure User Authentication FR-1:

The system shall allow for secure authentication of authorised SAID users. Users must be identified before they are allowed to use protected functionality or operational information. Authentication will determine the identity of the user and enable the system to determine the permissions for each role. Unsuccessful authentication shall not be allowed for users to enter into protected roster, staff, task, assignment or operational information.
The authentication shall be compliant to ensure that user credentials are safeguarded and not accessible to unauthorized users (Wright, 2025) . The computer's security is reinforced by role based access control. The system shall provide role based security, so that users shall be given the right to use only such functionality which is appropriate for their roles. Supervisors shall have access to the existing roster/force-management capabilities necessary to fulfill their operations. A primary access for staff will be to their own schedule, tasks, notifications to/from them relevant and/or allowed task-status functionality.
The system shall implement authorisation at the application level, meaning that certain parts of the functionality may not be accessible without the user bypassing the interface control.

Staff Profile and Workforce Information Management FR-2

The system shall store staff-related data necessary to facilitate the operation of the system and allocation of staff. The information in a staff profile includes the name of the employee, their operational role, operational area, skills applicable to their role, availability details, and active/inactive status. The kind of information stored should be kept to a minimum for the system's operational needs. Authorised users will have the ability to keep staff information if appropriate. Future scheduling activities should be updated as necessary when there are changes to the information of the relevant staff.

Operational Area Management FR-3

The system shall allow for configuration of operational areas to enable the association of tasks to the area in which they may be expected to occur.
Operational areas that have been identified through the existing roster include reception, pharmacy, wards, kennels, theatre, field outreach and equipment related areas. These areas should not be locked into the system since the operating environment for SAID may change. This system should therefore provide for authorised users to operate in operational areas and to have suitable relationships between areas and task assignments (Jama Software, 2021) .

Recurring Task Template Management FR-4

The system shall enable authorised supervisors to develop task templates that can be used for repeatable operational tasks.

A task template should include all the necessary information so that the task can be reproduced automatically. These can include task name, task description, operational area, task duration, frequency, priority, task skills, and if the task is critical.
The idea behind the use of reusable templates is to minimize repetitive manual data entry and ensure consistency when supervisors are creating repeat rosters (Jama Software, 2021).

Daily Roster Management FR-5

The system should enable supervisors who are authorised to make and keep daily rosters. Supervisors can set up duties, assign employees, adjust assignments, delete assignments (where applicable) and change operational schedules. Roster changes need to be reflected throughout the system to avoid the staff and supervisors seeing conflicting versions of the roster (santhosh, 2024) .

Manage the weekly roster.Manage Weekly Roster FR-6

The system shall depict the entire roster information to the supervisor on a weekly basis. The weekly view is important because there are often repetitive duties that need to be completed over a number of days that are included in SAID's duties. Supervisors must know what tasks are completed each day of the week, and not just every day individually.
The weekly view should enable supervisors to see staffing trends, overlapping duties, coverage gaps, and scheduling issues before individual days go live, however.Create and copy rosters, generate recurring schedules. The system to support creation of new rosters based on existing roster structures and task templates that can be reused.This feature is designed to minimize repetitive tasks involved in making schedules with recurring tasks.

If the previous roster has been duplicated, there should be a way for the supervisor to check and adjust the duplicated information before the new roster is issued. This way the re-use of recurring structures is possible, while making it possible to change structures in the course of a specific operational period.

Staff Assignment Management FR-7.

If a task is to be done by more than one employee, the system shall enable the authorised supervisor to assign one or more staff members to the task.
Assignments should be written in a way that would enable the staff member to know, upon receipt, what is expected of him. This includes the task, date, start time, end time, operational area, priority and any extra instructions.The system should keep staff assignments tied to the corresponding roster and task, allowing the supervisor(s) to track it during the operational period. The purpose of this form is to allocate staff to other locations.This form is for the staff re-assignment. The system shall provide for the re-allocation of operational duties, as and when necessary, by authorised supervisors (Microsoft.com, 2026). Staff shortage, change in staff availability, operational needs or change in proposed roster may necessitate reassignment.
If a reassignment causes a scheduling conflict or if it leaves another critical duty uncovered, the system should notify the supervisor and alert them to the scheduling conflict or duty void.

Roster Publication FR-8

Once the schedule is prepared and reviewed, the system shall provide authorised supervisors with the capability to publish a roster. The system needs to be able to tell the difference between information that is currently being prepared for the roster, and information that has been published for the staff to use (Harrison, 2026) .

If a roster is published, the appropriate assignments shall be published for the staff members who are impacted and notification shall be started, if notification requirements have been set. The Personal Staff Schedule is detailed here.Personal Staff Schedule: Detailed here. Staff will be provided with a personal timetable with staff duties. The schedule shall contain enough information for staff members to know what they are expected to do, when they need to do it, where they need to do it, and any other instructions that are related to the work that need to be followed. The staff facing interface should be designed with mobile communication in mind since the staff may need to access their schedules while walking between hospital, animal care and outreach locations (Harrison, 2026).

Task Status Management FR-9

Staff members will have the ability to mark their tasks as completed.
Version 1 will provide the following statuses: Scheduled, In Progress, Completed, and Unable to Complete. The Scheduled status shall be used to show a responsibility has been given, but no work has been started. The In Progress status shall be used to show that the employee has begun to work on the responsibility. Completed shall be used to show that the work assigned has been completed. Unable to Complete shall be used when the employee is unable to finish the Responsibility and shall include an explanation (where applicable). This will give supervisors a more accurate operational situation than a static roster that can compare planned responsibilities with the progress made.

Operational notification and alert FR-10

Significant operational events shall give relevant notifications to the system. Notifications will be used to inform about new rosters, assignment changes, reassignment of duties, critical duties approaching or overdue, urgent operational notifications, and unassigned critical duties. To avoid unnecessary communication and having to deal with notification overload, notifications shall be targeted to relevant users. The direction of notifications to changes or information for which users should be paying attention should be first through the roster and task information in the system (sqadmin, 2025) .

Identify scheduling conflict FR-11

The system shall detect possible time conflicts in staff assignments. A conflict can happen when a staff member is assigned to two or more jobs, or when an assignment conflicts with the relevant scheduling information. The system should alert the supervisor to a potential conflict, especially when the supervisor is generating, editing or posting a roster.

This capability is not intended to supplant the supervisor's operations sense, but rather to offer a mechanism to identify possible scheduling issues that may not otherwise be recognized when manually building the roster.

Uncovered Critical Duty Detection FR-12

The system shall be able to determine critical duties that have no staff member assigned. A critical duty is a duty of operation which SAID believes should be performed, and thus needs to be covered by a suitable workforce. If the duty is not assigned, the system should show the condition to the supervisor to allow for corrective action before the duty is due. The exact definition of a critical duty needs to be agreed to by SAID prior to implementation.

Supervisor Operational Dashboard FR-13

The system shall give supervisors a single view of the current condition of the system. The dashboard will possess the capabilities of displaying scheduled duties, current assignments, tasks in progress, completed tasks, outstanding tasks, unable to complete tasks, unassigned critical duties, scheduling conflicts, and areas of operation that need attention. The dashboard should focus on information that helps supervisors to make quick decisions on the operation. Interventions should be distinguishable from regular operational information.

Operational Notes FR-14

Authorised users should be able to add operational notes to tasks or rosters.
Notes can be used to add instructions, explain changes, record what is going on in operation and/or to send information to another authorised user. The responsibility described by the notes should be in an appropriate relationship with the system; such as, a note should not be separated from the operational context to which it applies. This will display the roster and task history (TeachMeSurgery, 2018). The system shall keep an adequate record of pertinent roster and task activity. History should be visible for events like creation of rosters, publication of rosters, changes in assignments, reassignments, changes in task status. This will enhance accountability of operations and enable authorised users to trace the history of a schedule/take and see how it has developed (TeachMeSurgery, 2018).

Roster Archiving FR-15

Completed or historical rosters shall be able to be archived in the system.
Archived rosters should be kept available for the appropriate operational use, but not be used as the active roster. This will ensure that there is no unnecessary impact on the current operational management due to existing schedules. However, the retention period and archival rules should be discussed with SAID prior to implementation.

Core Operational Workflow

The functional requirements are linked together in a central operation stream that models how the system can enable SAID's operations. Planning is the beginning point of the workflow (Luisi, 2014) . A supervisor can create a new roster or pick from existing roster structures and recurring task templates to define the planned responsibilities for the operational period for which the roster is created. Allocation is the next step, where staff are allocated to the roles listed in the roster. After the initial allocation is done, the system verifies its schedule, looking for possible conflicts and duties that have not been assigned. Once the schedule has been reviewed by the supervisor the roster can be published. Access to their personal schedules is then given to relevant personnel and notifications are made to appropriate personnel about their assignment.

Staff members complete the tasks assigned to them and mark them as completed as tasks progress. Supervisors are able to track these updates on the operational dashboard and take action when a task has not been completed, a duty has not been fulfilled, a scheduling issue has arisen or new operational need has emerged. Any relevant changes and actions are retained as operational history to assist with accountability and review of future operations (Luisi, 2014).

The overall operational workflow can thus be viewed as:
Plan → Allocate → Validate → Publish → Notify → Execute → Update → Monitor → Intervene → Record

This is a workflow example, showing that the proposed system is not just a digital roster-display application, but rather an operational-management system.

Requirements Outside Version 1

The Version 1 requirements are purposely confined to the operational coordination problem that was identified in the client engagement. The system shall NOT be used to keep animal medical records, veterinary diagnoses, treatment history, prescriptions, clinical notes or other clinical information. The roles are not part of the proposed platform and are still within the realm of the current clinical systems of SAID.

The system will not be replacing ezyVet or any clinical functionality that currently exists in ezyVet. Version 1 will not include direct integration with ezyVet unless SAID explicitly states that this integration is to be enhanced at a later date. It does not include financial functionality. Version 1 will not be able to handle accounting, payroll, invoicing, processing donations, point-of-sale operations or financial reporting. Initial scope does not include public facing services. It shall not offer Public Adoption functionality, lost and found reporting, public SOS reporting and general public volunteer management functionality. The system shall also not deliver AI-based veterinary diagnoses, treatment advice and clinical choices. Future AI capability would be subject to its own assessment, approval and would have to be consistent with the limits of the system.

9.1.6 User Roles

This section defines the people who will use the proposed SAID Operations Hospital Workforce, Roster and Task Management System and translates the operational evidence into user stories, a user-experience (UX) journey map and a work breakdown structure (WBS). The analysis responds to sections 9.1.6 and 9.1.7 of the module manual (IIE, 2026). It is based primarily on the client discovery meeting and the Friday hospital roster supplied by SAID (SAID, 2026a; SAID, 2026b).

The proposed solution is not intended to replace professional veterinary judgement. Rather, it is intended to improve the coordination, visibility and traceability of operational work. This distinction is important because the roster includes time-sensitive clinical support activities, ward responsibilities, theatre preparation, admissions, stock-related duties, equipment checks and outreach work. A system that represents these activities accurately can reduce avoidable doubt while preserving supervisors' authority to make context-sensitive decisions.

The design follows a human-centred approach. Human-centred design begins with an explicit understanding of users, tasks and context, involves users throughout development, and iteratively evaluates design solutions against user requirements (ISO, 2019; NIST, 2021). In this project, the client evidence therefore functions as the initial design baseline, while validation with Sally and representative staff remains necessary before the requirements are treated as final.

Operational Context and Design Principles

SAID provides professional veterinary care and animal-care education to pets and working animals in marginalized communities (SAID, n.d.). The proposed system must support a busy operational environment in which staff may move between reception, pharmacy, wards, theatre, stores, vehicles and field locations. The interface should be usable under time pressure, on different device types and by users with different levels of digital confidence.

The following principles guide the requirements in this section:

Clarity: show the duty, time, location, responsible person and current status without requiring users to interpret the complete roster.
Role-appropriate access: give supervisors coordination and monitoring functions while giving staff a focused view of their own responsibilities.
Error prevention: identify overlapping assignments, unassigned critical duties and incomplete information before a roster is published.
Visibility and accountability: preserve status changes, notifications and roster history so that operational decisions can be reviewed.

Human-centred iteration: validate prototypes with the client and representative users, then refine the design based on observed difficulties and feedback (ISO, 2019; NIST, 2021).

User Roles

There are two roles in the system: Supervisor/Administrator and Staff Member. This is because we found out that there are two supervisors who make and change the work schedule and a lot of staff members who do the tasks they are given. The information we found shows that two supervisors are in charge of making sure the work schedule is right, and the staff members do their jobs (SAID, 2026a). Reception, pharmacy, ward care, theatre support and field outreach are treated as duty or skill categories within the Staff Member role rather than as separate system roles. This reduces unnecessary complexity in the first release, although permissions for clinical and non-clinical staff should be confirmed during requirements validation.

Role	**Purpose**	**Responsibilities**	**Primary interface**
Supervisor/Administrator	Coordinates daily hospital operations and maintains the authoritative roster.	Create, edit, copy, publish and archive rosters; assign and reassign duties; manage recurring templates; monitor progress; resolve conflicts; issue updates; and review historical records.	Web/desktop dashboard, roster editor, alerts, monitoring dashboard and reporting/history views.

Staff Member	Executes duties assigned through the roster, whether in the hospital or on field outreach.	View assigned duties; inspect time, location and instructions; acknowledge changes; update task status; report inability to complete a duty; and review recent duty history.	Mobile-first schedule, task detail screen, notifications and status controls.

Supervisor/Administrator

The person in charge or the person who manages the team is the one who makes things run smoothly. This job matches the two supervisors mentioned in the discovery evidence, who now create and change schedules by hand (SAID, 2026a). The proposed system should support both planned scheduling and rapid intervention. For example, a supervisor may need to respond to staff absence, an unavailable field vehicle, a change in patient demand or an uncovered duty. The interface should therefore make exceptions visible and allow controlled reassignment without requiring the entire roster to be recreated. Supervisor permissions should be designed according to least privilege: users should have enough access to coordinate work, but sensitive settings and account administration should be limited to authorised administrators. The exact permission boundary should be confirmed with SAID before implementation.

Staff Member

The Staff Member role represents the broader workforce. The Friday roster identifies responsibilities across reception, pharmacy, food and donation storage, vehicle and equipment checks, patient admissions, General, Cat, Ortho, ICU, Parvo and Semi-Isolation wards, theatre support. (SAID, 2026b).

The staff experience should be deliberately focused. Instead of presenting the entire operational plan, the mobile interface should prioritize the user’s next duty, today’s schedule, location, timing, instructions and any changes issued by a supervisor. This supports suitability for the task and reduces cognitive load, which are central considerations in human-centred interaction design (NIST, 2021).

9.1.7 User Stories

User Stories and Acceptance Criteria

The stories below use the standard “As a user, I want…, so that…” structure required by the module manual (IIE, 2026). Acceptance criteria have been added to make each story testable. The criteria are proposed system behavior and should be validated with SAID before development.

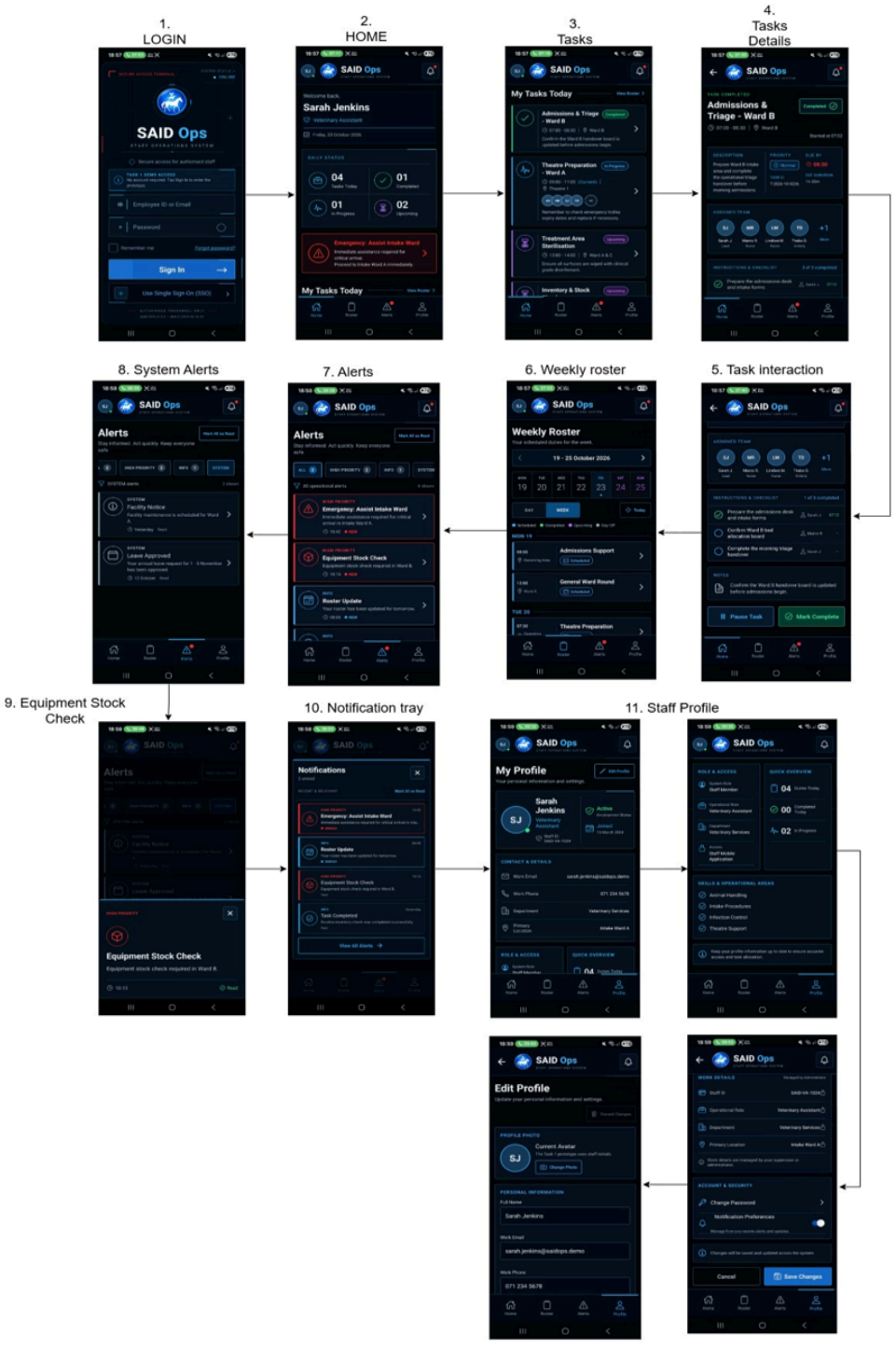
ID	Feature	Role	User story	Rationale/evidence	Proposed acceptance criteria
US-01	Roster creation and publication	Supervisor	As a Supervisor, I want to create and publish rosters so that staff duties are clear.	Staff duties are defined before work begins and the manual process is reduced (SAID, 2026a).	The supervisor can select a date range; required fields are validated; the roster remains a draft until published; publication records the date, time and publishing user.
US-02	Roster duplication	Supervisor	As a Supervisor, I want to copy previous rosters so that recurring duties are not re-entered.	Recurring patterns such as pharmacy duties and stock takes are not re-entered unnecessarily (SAID, 2026b).	The supervisor can select a source roster; dates are shifted appropriately; copied duties can be edited before publication; the original remains unchanged.
US-03	Personal schedule	Staff Member	As a Staff Member, I want to view my daily duties so that I understand my responsibilities.	I can understand my responsibilities without interpreting the complete roster.	The screen shows assigned duties in time order; unassigned users cannot view restricted staff details; the user can open duty details.
US-04	Shared duty assignment	Supervisor	As a Supervisor, I want to assign staff to duties so that shared work is coordinated.	Shared responsibilities, including theatre preparation, are represented accurately (SAID, 2026b).	The supervisor can add multiple staff; each assignment is visible to the relevant staff; duplicate assignment warnings are shown where applicable.
US-05	Reassignment	Supervisor	As a Supervisor, I want to reassign duties so that operational gaps are addressed.	Operational gaps can be addressed without rebuilding the roster (SAID, 2026b).	The supervisor can change the assignee; the change is logged; affected staff receive a notification; the updated roster is visible immediately.
US-06	Duty detail	Staff Member	As a Staff Member, I want to view duty details so that I can prepare properly.	I can prepare for time-critical work, including theatre preparation beginning at 09:30 (SAID, 2026b).	Duty details include time, location/area, description, collaborators and relevant notes; times are displayed consistently.
US-07	Recurring templates	Supervisor	As a Supervisor, I want to create recurring-duty templates so that routine tasks are not omitted.	Routine responsibilities such as vehicle checks and oxygen-cylinder inspections are not omitted (SAID, 2026b).	A template stores duty name, frequency, expected duration, skills and default area; the supervisor can activate, edit or deactivate it.

US-08	Task status	Staff Member	As a Staff Member, I want to update duty statuses so that supervisors can track progress.	Supervisors have current visibility of operational progress.	Only permitted status transitions are available; completion time and user are recorded; unable-to-complete requires a reason or comment.
US-09	Monitoring dashboard	Supervisor	As a Supervisor, I want to monitor all duties from one dashboard so that unattended work is identified.	Unattended responsibilities can be identified early.	The dashboard filters by date, area, staff member and status; overdue and uncovered duties are visually distinct; selecting a duty opens its details.
US-10	Critical-duty warning	Supervisor	As a Supervisor, I want warnings for unassigned critical duties so that important work is covered.	Admissions, ICU rounds and similar responsibilities are not left uncovered.	Critical duties are configurable; a draft cannot be published, or requires explicit override, when a critical duty is uncovered.
US-11	Conflict warning	Supervisor	As a Supervisor, I want warnings for overlapping assignments so that scheduling conflicts are prevented.	Conflicts are resolved before publication.	The system checks time overlaps and unavailable staff; the warning identifies the conflicting duties; the supervisor can resolve or document an override.
US-12	Change notification	Staff Member	As a Staff Member, I want notifications about duty changes so that I remain informed	I am informed without repeatedly checking the roster.	The notification identifies what changed, when it applies and who issued it; the updated duty appears in the personal schedule.
US-13	Field outreach	Supervisor	As a Supervisor, I want to schedule outreach teams by area so that field activities are coordinated.	Groupings such as Olievenhoutbosch/Diepsloot and Tembisa/Ivory Park are clearly trackable (SAID, 2026b).	The supervisor can record area, team, departure time, vehicle, equipment and status; field duties appear in both supervisor and staff views.
US-14	Field preparation	Staff Member	As a Staff Member, I want to view outreach details so that I can prepare for departure.	I can prepare before departure.	The duty identifies route/area, team members, vehicle, equipment checklist and relevant instructions.
US-15	Roster history	Supervisor	As a Supervisor, I want to review past rosters so that staffing patterns can be identified.	Recurring staffing gaps and operational patterns can be identified.	Historical rosters are searchable; records are read-only by default; filters support date, area, duty and completion status.
US-16	Duty history	Staff Member	As a Staff Member, I want to view my duty history so that I can track completed work.	I have a record of complete responsibilities.	The history shows date, duty, area, status and completion time, subject to SAID's retention and privacy rules.

UX Journey Map

A journey map visualises the process a person follows to accomplish a goal (Nielsen Norman Group, 2023). The following maps describe the intended screens, user actions, system response, possible pain points and design opportunities. They complement the supplied diagrams and provide textual traceability for prototype review. The exact screen names should be aligned with the final prototype.

Mobile Prototype: Figure X


1. LOGIN
2. HOME
3. Tasks
4. Tasks Details
5. Task interaction
6. Weekly roster
7. Alerts
8. System Alerts
9. Equipment Stock Check
10. Notification tray
11. Staff Profile
SAID Ops
Sign In
Sarah Jenkins
My Tasks Today
Admissions & Triage - Ward B
Weekly Roster
Alerts
Notifications
My Profile
Edit Profile
Equipment Stock Check

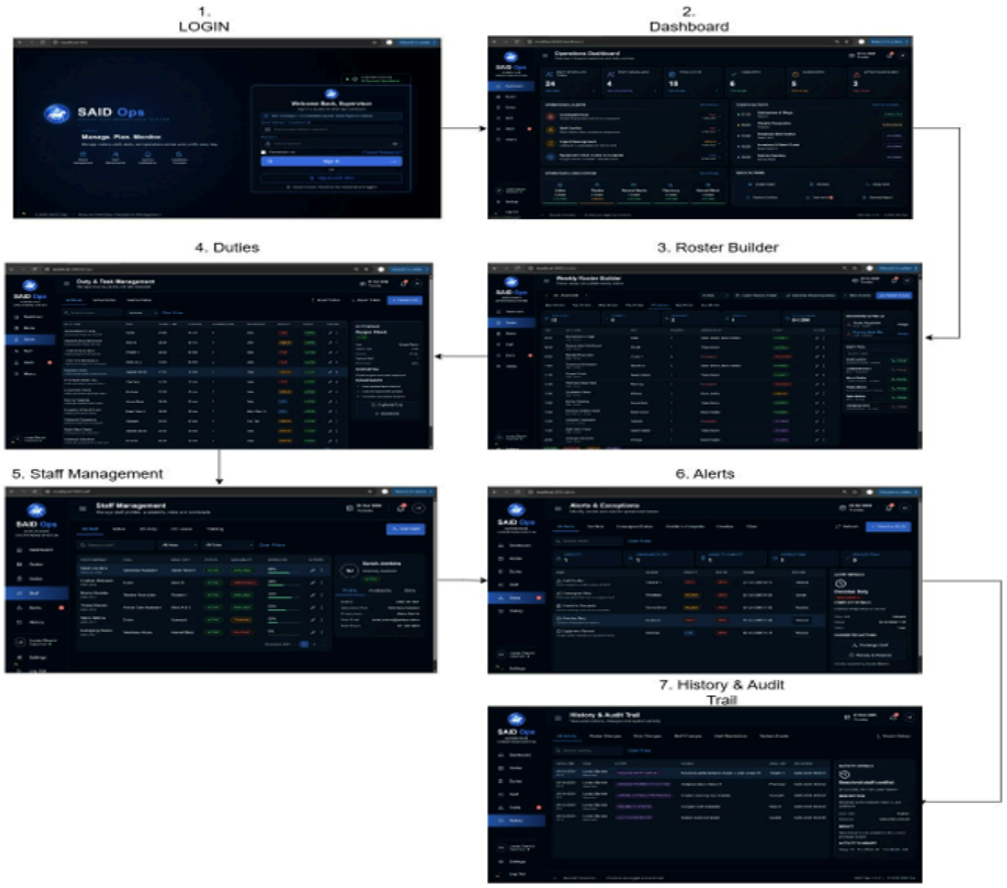
Staff Member Journey: Mobile Prototype

The Staff Member journey is based on the completed Task 1 mobile prototype, developed in Expo/React Native using shared local dummy data. The prototype is frontend-only, and Sarah Jenkins, Veterinary Assistant, is a fictional example user. The complete navigation sequence is presented in Figure X.

After signing in with an employee ID or email, the user accesses the home screen, which displays a daily status summary, emergency alerts and assigned tasks. The bottom navigation provides access to Tasks, Roster, Alerts and Profile. Users can view task details, checklists, weekly duties, operational alerts and personal information. The header bell icon opens the Notification Tray, while the Edit Profile screen allows permitted updates to contact details, profile photo, password and notification preferences.

A likely pain point is that staff members working across multiple duty categories (e.g. ward care and theatre support on the same day) may struggle to judge priority when several tasks appear together on that screen. The Task Details screen addresses this by surfacing priority and a checklist, giving staff a clear basis for sequencing their work rather than relying on duty order alone.

Web Prototype: Figure Y



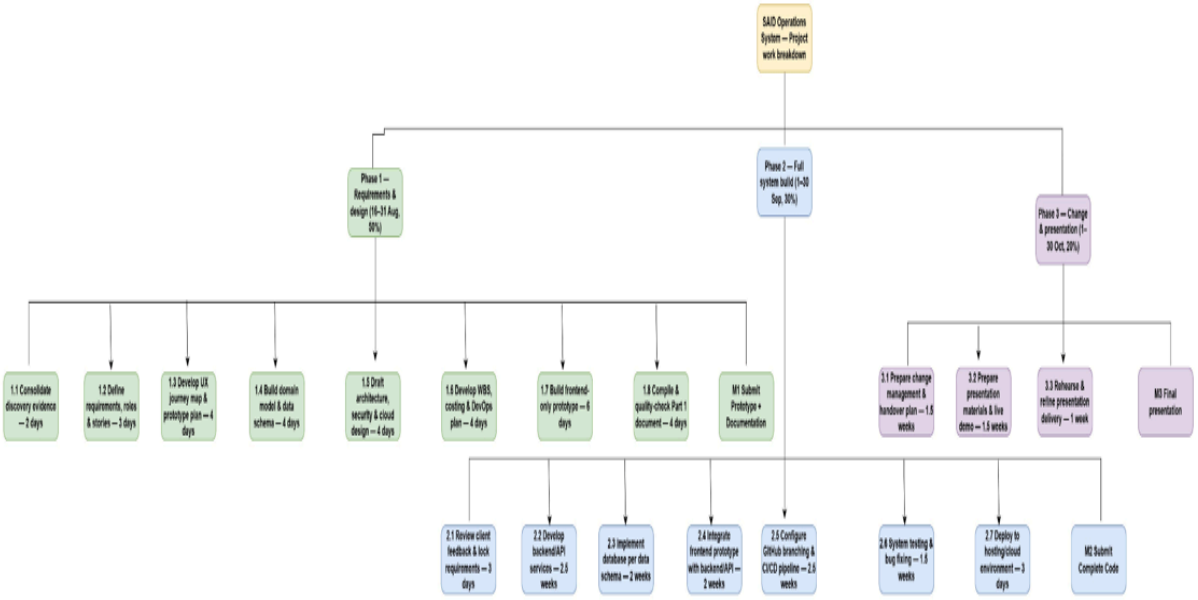
Supervisor/Administrator Journey: Web Prototype

The Supervisor/Administrator journey is based on the completed Task 1 web prototype, which uses frontend only. Lerato Dlamini, Supervisor, is a fictional example user. The complete navigation sequence is presented in Figure Y.

After signing in, the supervisor accesses the Dashboard, which displays scheduled duties, task progress, alerts and area status. The sidebar provides access to the Roster Builder, Duties, Staff and Alerts screens. The supervisor can create and update rosters, assign staff, manage duties, review staff information, resolve conflicts and monitor critical unassigned duties. Resolved actions are recorded in the History and Audit Trail.

The main workflow is Login → Dashboard → Roster Builder → Alerts → History. The Duties and Staff screens are reference areas available through the persistent sidebar. The web and mobile prototypes are conceptually linked but are not yet technically integrated. Backend integration and end-to-end validation are planned for Phase 2.

A likely pain point for supervisors is losing track of which conflicts or critical-duty warnings have already been actioned during a busy publishing cycle. The Alerts screen's categorisation, combined with the History and Audit Trail, gives supervisors a way to confirm that a warning was resolved rather than simply dismissed, reducing the risk of an unresolved conflict being missed before publication.


SAID Operations System — Project work breakdown
Phase 1 — Requirements & design (16–31 Aug, 50%)
Phase 2 — Full system build (1–30 Sep, 30%)
Phase 3 — Change & presentation (1–30 Oct, 20%)
1.1 Consolidate discovery evidence — 2 days
1.2 Define requirements, roles & stories — 3 days
1.3 Develop UX journey map & prototype plan — 4 days
1.4 Build domain model & data schema — 4 days
1.5 Draft architecture, security & cloud design — 4 days
1.6 Develop WBS, costing & DevOps plan — 4 days
1.7 Build frontend-only prototype — 6 days
1.8 Compile & quality-check Part 1 document — 4 days
M1 Submit Prototype + Documentation
2.1 Review client feedback & lock requirements — 3 days
2.2 Develop backend/API services — 2.5 weeks
2.3 Implement database per data schema — 2 weeks
2.4 Integrate frontend prototype with backend/API — 2 weeks
2.5 Configure GitHub branching & CI/CD pipeline — 2.5 weeks
2.6 System testing & bug fixing — 1.5 weeks
2.7 Deploy to hosting/cloud environment — 3 days
M2 Submit Complete Code
3.1 Prepare change management & handover plan — 1.5 weeks
3.2 Prepare presentation materials & live demo — 1.5 weeks
3.3 Rehearse & refine presentation delivery — 1 week
M3 Final presentation

Work Breakdown Structure

WBS Tree Diagram for SAID Operations System

A WBS decomposes project scope into manageable, deliverable-oriented work packages. The schedule below retains the three assessed milestones and clarifies dependencies, ownership and resources. Durations are estimates for planning rather than guarantees; they should be reviewed after feedback and risk assessment (Project Management Institute, 2021).

	Task	Description	Duration	Predecessors	Responsible member(s)	Resources
1.1	Consolidate discovery evidence	Organize meeting notes and Friday roster into a traceable evidence base.	2 days	—	Nyiko Masinge	Meeting notes; Roster.xlsx; laptop
1.2	Define requirements, roles and stories	Draft functional/non-functional requirements and evidence-based user stories.	3 days	1.1	Nyiko Masinge; Maseo Mandiwana	Module manual; shared document
1.3	Develop UX journey map and prototype plan	Map screen-level navigation for supervisor web and staff mobile experiences.	4 days	1.1	Maseo Mandiwana	draw.io; Stitch account
1.4	Build domain models and schema	Design entities and relationships for rosters, duties, staff and notifications.	4 days	1.2	Tival Koovarjee	ERD tool; roster data
1.5	Draft architecture, security and cloud design	Define architecture, patterns, hosting approach and security controls.	4 days	1.2	Ndivhuwo Ndou	Diagramming tool; cloud documentation
1.6	Develop WBS, costing and DevOps outline	Prepare plan, cost assumptions and CI/CD approach.	4 days	1.1	Maseo Mandiwana; Aaryan Makan; Ndivhuwo Ndou	Excel/Word; GitHub
1.7	Build front-end-only interactive prototype	Develop clickable supervisor and staff interfaces using representative data.	6 days	1.3, 1.4, 1.5	Aaryan Makan; Maseo Mandiwana	Stitch/Figma; laptop
1.8	Compile and quality-check Part 1	Merge sections, align terminology, test traceability and check Harvard referencing.	4 days	1.2, 1.4, 1.5, 1.6, 1.7	Whole group	Shared document; referencing guide
M1	Submit prototype and documentation	Formal Part 1 submission.	—	1.8	Whole group	IIE submission portal

Phase 1: Requirements, Design and Prototype (16–31 August 2026; Prototype and Documentation, 50%). The short period makes early parallel work important. Tasks 1.2–1.6 can proceed in parallel after evidence consolidation, while prototype construction should use the validated journey map and data assumptions.

	Task	Description	Duration	Predecessors	Responsible member(s)	Resources
2.1	Review feedback and lock requirements	Incorporate Part 1 feedback and establish the development baseline.	3 days	M1	Nyiko Masinge	Feedback notes; requirements document
2.2	Develop backend/API services	Build services for rosters, duties, assignments, status and notifications.	2.5 weeks	2.1	Ndivhuwo Ndou; Tival Koovarjee	Backend framework; IDE; cloud account
2.3	Implement database	Implement structures defined in the data schema.	2 weeks	2.1	Tival Koovarjee	DBMS; schema documentation
2.4	Integrate frontend with backend	Replace static data with live API-supported workflows.	2 weeks	2.2, 2.3	Aaryan Makan; Ndivhuwo Ndou	API documentation; Postman
2.5	Configure GitHub and CI/CD	Implement branching, automated build and deployment pipeline.	2.5 weeks	2.1	Ndivhuwo Ndou	GitHub Actions; repository access
2.6	System testing and bug fixing	Test core workflows, accessibility/usability risks and defects.	1.5 weeks	2.3, 2.4, 2.5	Whole group	Test cases; bug tracker
2.7	Deploying to hosting/cloud	Deploy and smoke-test the system in the selected environment.	3 days	2.6	Ndivhuwo Ndou	Hosting account; deployment scripts
M2	Submit complete code	Formal working-system submission.	—	2.7	Whole group	IIE portal; GitHub repository

Phase 2: Full System Development (1–30 September 2026; Complete Code, 30%). Development should begin only after the requirements baseline is agreed. Database and API work should be coordinated closely because the user stories depend on consistent identifiers for users, duties, rosters, assignments and notifications.

	Task	Description	Duration	Predecessors	Responsible member(s)	Resources
3.1	Prepare change management and handover plan	Document rollout, training, support and handover activities.	1.5 weeks	M2	Aaryan Makan	Change-management template; Word
3.2	Prepare presentation and live demonstration	Develop slides and a scenario-based walkthrough.	1.5 weeks	M2	Whole group	PowerPoint; working system
3.3	Rehearse and refine delivery	Rehearse timing, content, transitions and role allocation.	1 week	3.2	Whole group	Slides; meeting/Teams call
M3	Final presentation	Formal presentation to markers and client.	—	3.1, 3.3	Whole group	Presentation venue/platform

Phase 3: Change Management and Presentation (1–30 October 2026; Presentation, 20%). The change-management plan should address user orientation, communication, support during transition and feedback after initial use. A technically functional system may still fail to deliver value if staff do not understand when and how it should be used.

Traceability, Validation and Risk Controls

The proposed roles and stories are traceable to the discovery evidence. The Supervisor/Administrator role is supported by the description of two supervisors who maintain the roster (SAID, 2026a). Staff stories are supported by the duty categories, timings, locations and team groupings contained in the Friday roster (SAID, 2026b). The design rationale is further supported by human-centred design guidance, which recommends understanding context, specifying requirements, producing design solutions and evaluating those solutions iteratively (ISO, 2019; NIST, 2021).

Evidence/problem	Related stories (ID)	Design response	Risk addressed
Manual roster preparation	US-01, US-02, US-15	Roster editor; templates; history	Time required to create and review rosters
Multiple duty categories and areas	US-03, US-06, US-13, US-14	Personal schedule; duty details; field module	Incorrect interpretation of responsibility
Time-critical theatre work	US-06, US-10, US-11	Time display; critical-duty and conflict validation	Late or uncovered duty
Vehicle/equipment constraints	US-05, US-13, US-14	Resource fields; reassignment; checklist	Outreach disruption
Need for operational visibility	US-08, US-09, US-12	Status dashboard; notifications	Supervisors lack current information

Before development is completed, the group should conduct a validation session with Sally and, where possible, representative users from both role categories. The session should confirm role boundaries, critical duties, terminology, notification expectations, mobile-device constraints, field connectivity, data retention and the consequences of an “unable to complete” status. Findings should be recorded as approved requirements, amendments or unresolved questions.

Two stories are not yet represented by a dedicated screen in the current front-end-only prototype: US-14 (outreach and vehicle/equipment preparation) and US-16 (staff duty history). Both remain valid requirements supported by the discovery evidence, but no corresponding screen exists in the confirmed mobile capability set (Login, Home, Tasks, Task Details, Weekly Roster, Alerts, Notification Tray, Profile). These are deferred to Phase 2, when backend integration will allow historical records and field-outreach data to be modelled and displayed.

2.8 Conclusion

This section presents a focused role model, evidence-based user stories with acceptance criteria, confirmed UX journey maps for the Supervisor/Administrator and Staff Member roles, and a milestone-aligned work breakdown structure for the proposed SAID operations system. Unlike the initial draft, the journey maps are based on the completed Task 1 web and mobile prototypes, providing markers and the client with a practical basis for review.

The design prioritises clarity, appropriate role access, conflict prevention, real-time visibility and historical accountability, in accordance with human-centred design principles (ISO, 2019; NIST, 2021). Two matters require confirmation before finalisation. The exact structure of the Profile and Edit Profile sections should be verified with the prototype developers. In addition, the web and mobile prototypes currently operate independently without a shared backend. End-to-end validation of supervisor actions and staff updates should therefore be completed during Phase 2. The next step is to conduct a validation session with Sally and representative staff members to confirm role boundaries, terminology and operational assumptions before backend development and integration begin.

9.1.8 Non-functional requirements

Non-Functional Requirements Overview

Functional requirements describe what the SAID Operations must be able to do and the non-functional requirements describe the attributes of the system that determine its ability to perform the functional requirements effectively, securely, reliably, and appropriately (Jama Software, 2021) .

The significance of these requirements is more important because the proposed system would be used to help the real operation of the activities. If the system becomes difficult to use, unreliable, insecure, slow, or unavailable when they need it, it may create more operational challenges than what they were originally looking to solve when the client was originally engaged.

The non-functional requirements hence cover aspects of security, performance, usability, mobile access, reliability, availability, maintainability, scalability, data integrity, auditability, privacy, compatibility, backup and recovery, logging, as well as notification reliability.

Security Requirements

Secure Authentication NFR-1

Access to protected functionality shall be the responsibility of the system and require authentication. Users must be able to use staff schedules, staff assignments, operational information, and administrative functionality only after successfully authenticating. Authentication credentials shall be stored in a secure manner and never in plain text (Jama Software, 2021) .

Role-Based Authorisation NFR-2

The system shall maintain role-based authorisation for the purpose that a user can only access functions that are relevant to their role. It shall not be possible for a staff member to create, modify, publish or manage rosters for a user when the user does not have the permission to perform those activities in their assigned role (Singh, 2026) . You cannot have interface restrictions alone as authorisation needs to be done at the server or application-service level. This NFR sets out to safeguard staff and operational data. Unauthorised access to staff related information, and operational information shall be prevented. The system shall adhere to the principle of least privilege: systems shall be designed in a way that the information and permissions a user needs to do his/her job are only those that are necessary.

Where personal information is being processed, the implementation should consider relevant South African privacy and data protection principles, such as the Protection of Personal Information Act (where applicable).

Password Security NFR-3

If a password-based authentication method is used, passwords should be securely hashed with an appropriate modern password-hashing mechanism. Plaintext passwords should not be stored in the application's database or configuration.

Input Validation NFR-4

All data provided by the users will be checked for accuracy before being used or recorded. Staff information, task information, dates, times, assignments, notes, roster information, and other user-controlled information will be validated.
Client-side validation can be applied to enhance usability, but server-side validation shall be the authoritative validation.

Protection Against Common Application Attacks (CERT) NFR-5

The application shall have suitable protections against typical Web-application attacks. This shall encompass appropriate countermeasures for SQL Injection, Cross Site Scripting, Unauthorised access, Malicious input and any other application security risks. If applicable, the system shall also provide the following protection: Cross-site request forgery, brute-force authentication, insecure API use and session vulnerabilities. Objective: Secure Configuration Management.Goal: Secure Configuration Management. The configuration information shall not be hard coded in the application source code if it is sensitive. Environment variables and/or any secrets management system should be used to store database credentials, authentication secrets, API keys and/or tokens. Configuration files should not be checked into source-control repositories if they are sensitive (HCL SW Blogs, 2025) .

Performance Requirements

Application Responsiveness NFR-6

The system shall deliver a responsive experience upon common operations like viewing schedules, opening task details, updating status of tasks and accessing notifications. Finally, the end target for the performance should be validated throughout the implementation and testing phase depending on the infrastructure chosen for deployment.

Roster Performance NFR-7

The system shall be efficient to show information of daily and weekly roster regardless of the number of duties represented in the schedule. The 107 entries on the supplied Friday roster represent an operational load that this system provides - and with a similar load, should be able to do a comparable job without requiring significant degradation in usability or responsiveness.

Dashboard Performance NFR-8

Supervisors should be able to load information over the operation quickly and without doing too many operations on the database.
Information that is relevant to the immediate action of the operator, especially scheduling conflicts, unassigned critical duties, outstanding tasks, current assignments, and other factors that need supervision should be emphasized in the dashboard.

Usability Requirements

Ease of Use NFR-9

The system shall be easily figured out by its supervisors and employees without heavy technical background. Functionality of the interface focuses on a task's efficiency and simplicity of execution rather than being unnecessarily complex (IxDF - Interaction Design Foundation, 2016) .

NFR-10: Staff Usability

The Staff interface shall enable employees to easily identify what they are expected to do, by when this responsibility should be carried out, where the responsibility should be carried out and the status of the current task. Simple tasks like starting a task, completing a task or reporting that they can't complete a task should require limited interaction (IxDF - Interaction Design Foundation, 2016) .

NFR-011: Supervisor Usability

The supervisor interface shall enable authorised personnel to easily identify operational conditions which require attention. Calendars, uncovered critical duties, outstanding tasks and other operational risks should be presented in a way that enables supervisors to be aware of the situation and take appropriate action (IxDF - Interaction Design Foundation, 2016) .

NFR-012: Interface Consistency

The system shall have uniform terms, navigations, buttons, status indicators, forms and interaction patterns. The meaning of task statuses should be uniform across the system to help staff and supervisors visualize what is happening during operations.

Mobile Accessibility and Responsive Design Requirements

NFR-013: Mobile Accessibility

A modern mobile device shall be able to access the staff facing interface.
This is to align with the working environment employees may have to travel between hospital, animal-care, and outreach provision environments to undertake duties.
Staff members should not be required to log onto a desktop computer just to see what their next task is or what the status of a task is.

NFR-014: Responsive Interface

The system shall automatically adjust itself to various screen sizes.
The supervisor interface might be optimised for desktop and laptop screens due to the fact that managing rosters involves presenting more information. Staff interface – must be mobile friendly, and information about tasks must be available quickly.

NFR-015: Accessibility

Accessibility should be included in the design with readable typography, sufficient contrast, meaningful labels, usable forms, understandable error messages, and usable keyboard navigation. If information is to be provided, it should not be presented in colour as this does not allow users to understand the information without a visual impairment or in poor lighting conditions (IxDF - Interaction Design Foundation, 2016) .

Reliability Requirements

NFR-016: Operational Reliability

While conducting normal operation, the system shall be capable of reliably accessing roster and task information. Sudden application errors should not result in silent loss of user changes. If an operation can not be completed, the system should let the user know why.

NFR-017: Error Recovery

Application, database and network errors shall be dealt with gracefully. Users must be given understandable feedback if the operation is not possible, and information about technical errors must be logged, but not made available to the users.

NFR-018: Data Integrity

Information related to rosters, assignments, tasks, staff, and operational areas shall be maintained. Invalid relationships should not be created. Rosters should not include staff invalid relationships, for instance, and task assignments should not refer to a non-existent staff member.

Availability Requirements

NFR-019: Operational Availability

The system must be available when the SAID supervisors and staff need operational information. It is important to verify the availability target through SAID based on availability requirements, deployment infrastructure and business requirements of the organisation. This course covers the principles of service failure handling. If the dependent service is temporarily unavailable there should be an appropriate error message. The system should not show information that implies that an unsuccessful operation was accomplished successfully. This is especially crucial in the context of roster changes and task-status updates as incorrect confirmation may result in erroneous operational data.

Maintainability Requirements

Maintainable System Architecture NFR-020.

Systems shall be developed in a modular structure which separates major system responsibilities. Implementation should be done using appropriate software engineering principles such as separation of concerns, modularity, reusability of components, appropriate abstraction and clear naming. This will enable the future developers to easily understand and alter one aspect of the system without impacting unrelated functions.

Technical Documentation NFR-021

The important system functionality, configuration requirements, database structure, application interfaces, deployment procedures and operational processes shall be documented. Documentation must be comprehensive enough for future development or authorised technical personnel to be able to understand how the system works and how it should be maintained.

Version Control NFR-022

The source code will be versioned with appropriate version control systems like git. Changes to the development should be recorded by the commit or other version-control history. Any sensitive configuration information should not be placed into source control.

Scalability Requirements

Workforce Scalability NFR-023

The system must be able to accommodate other SAID users when the needs of the organisation evolve. The architecture should not make any assumptions that limit the system to the first set of users considered in the planning of Version 1.

Operational Scalability NFR-024
The system should accommodate expansion of staff, geographic locations, repeated tasks, schedules, assignments and previous data without the need to redesign the system.
This is especially important as the supplied roster is quite large and future operational areas/duties may be added.

Feature Scalability NFR-025

The system architecture should be designed to enable additional functionality to be added to it without greatly affecting Version 1 functionality. Enhancements would include better reporting, operational analytics, intelligent scheduling, more notification options or approval integration with external systems. These are features that may be in the future but should not be needlessly stifled by the first design.

Requirements for Data Integrity and Auditability

Operational History NFR-026

Important roster, assignment and task data is to be traceable for changes.
System to keep events like roster creation, roster publication, assignment changes, reassignment and task status changes in the history.

Audit Information NFR-027

Historical operational records should indicate the user who made a change, the date and time at which the change was made, the roster or task affected, the type of action taken, and the previous and new values (if applicable) when the change was made. This will help with operational accountability and give supervisors more insight into the times that operational schedules have changed throughout time.

Data Collection is only required to the extent necessary to complete the NOC.
The system shall be used to collect only the information that is essential for the operation purposes outlined in the project. Do not gather information that is not specifically related to workforce scheduling, task allocation, communication or operational management just because it could be useful in the future.

Data Retention NFR-028

SAID will establish suitable retention periods for staff information, roster information, task information, operational notes and historical records. Until such periods are officially recognized as retention periods, retention requirements shall be deemed provisional.

Compatibility Requirements

Browser Compatibility NFR-029
The web application should work on the most popular browsers. Browser-support requirements should be decided at implementation and testing to ensure the application is deployed to compatible environments of the browsers available.

Device Compatibility NFR-030

The staff interface should be usable on the most up-to-date smartphones and tablet sizes. Supervisors might have to deal with more roster data, and the supervisor interface should be effective on normal desktop and laptop displays. The procedures for backing up and recovering the database.Database backup and recovery requirements.

Data backup operation NFR-031

Operational information should be backed up as per the schedule agreed upon considering the importance of SAID's rosters and tasks. The best backup frequency depends on the infrastructure, operations, volume of data and acceptable recovery goals of SAID.

Data Recovery NFR-032

The system must have a clear recovery procedure to recover operational data in the event of a database failure, infrastructure failure or other major system event.
Document and test recovery procedures prior to using it as a production operational platform.

Logging and Monitoring Requirements.

Application Logging NFR-033

The system should capture important technical and operational events that are necessary for troubleshooting, monitoring and accountability. Depending on the final implementation, these events may include application errors, authentication events, roster changes, assignment changes, and task-status changes.

Protecting sensitive information in logs NFR-032

Application logs shall not contain passwords, authentication tokens, API credentials or other sensitive security information. Logging should also be kept to a minimum of unnecessary exposure of personal information and should be based on the principle of recording only information essential to troubleshooting, monitoring and accountability.

Notification Reliability Requirements

Notification Reliability NFR-032

Notices of critical events and conditions to be communicated to others should be reliably delivered to the appropriate recipients. If notification delivery is not guaranteed, the system should either leave a notification in a suitable state or leave an error state, for investigation. The notification system shouldn't be used to replace the roster. The roster and task information in SAID Operations shall be the key operational source of truth and it is through the notifications that changes, urgent information or responsibilities that require action are drawn to attention.

REQUIREMENTS TRACEABILITY AND VALIDATION

Relating the Operational problem to the Proposed system

The requirements in this document state explicit links between SAID's operational problem and the proposed system functionality. The current operational issue is focused on coordinating people, places, schedules, tasks, task assignments, changes and task progress

(Weyers, 2022) . The proposed system tackles these areas by providing an integrated operational process, instead of adding each requirement as a standalone software feature. The relationship begins with the current operational challenge of manually managing an intricate workforce schedule. This leads to a business requirement for operational visibility across the board. This is addressed by the system's centralised roster management and task management (Weyers, 2022) . Then can come the roster-management feature, which serves as the basis for assigning staff and executing tasks. The conflict detection and uncovered duty identification feature provide an extra layer of operational control, allowing supervisors to more easily see potential issues ahead of their effects on operations.
Staff are sent their individual timetables and notifications once the timetable is made public. The dashboard and the operational history give supervisors information and accountability about the progress of the tasks, and the task status updates give information about the actual progress.

The relationship that comes out is therefore:

Operational Problem → Business Need → System Requirement → System Functionality → Operational Control → Staff Execution → Supervisor Oversight

This traceability guarantees that the proposed functionality is still linked to the operational requirements which were identified by SAID.

Requirements Validation

Requirements outlined in this section are based on information provided at initial engagement with clients and provided operational roster. These requirements will serve as a roadmap to develop Version 1, but some additional business rules need to be validated before the final development is complete (Jama Software, 2026) . These include, among others, the exact definition of a critical duty, the rules for staff availability, the skills required for specific duties, who is able to publish rosters, who is able to create urgent tasks, the channels of communication that are required, the time duration for which operational history is retained, and the precise requirements for privacy of staff information.

CONCLUSION

The SAID Operations project is a response to a specific operational need in the Society for Animals in Distress (SAID). Necessitates multiple staff roles, operations and time frames to be synchronised in its hospital, animal-care, outreach and supporting activities. The list of activities that can be run on Friday is quite extensive and complex as illustrated by the provided Fridays Roster which contains 107 activities. With all these responsibilities, a rigid work schedule is not enough to ensure good workforce coordination.

The proposed SAID Operations system will enable supervisors to create, maintain, validate, publish, modify and monitor operational rosters and staff to access and manage their assigned responsibilities in a centralised operational platform. The Version 1 scope deliberately focuses on workforce and operational coordination. The system will handle the people, operational areas, schedules, tasks, assignments, notifications, task progress and

operational history, as well as being separated from the clinical animal-record functionality provided by ezyVet.

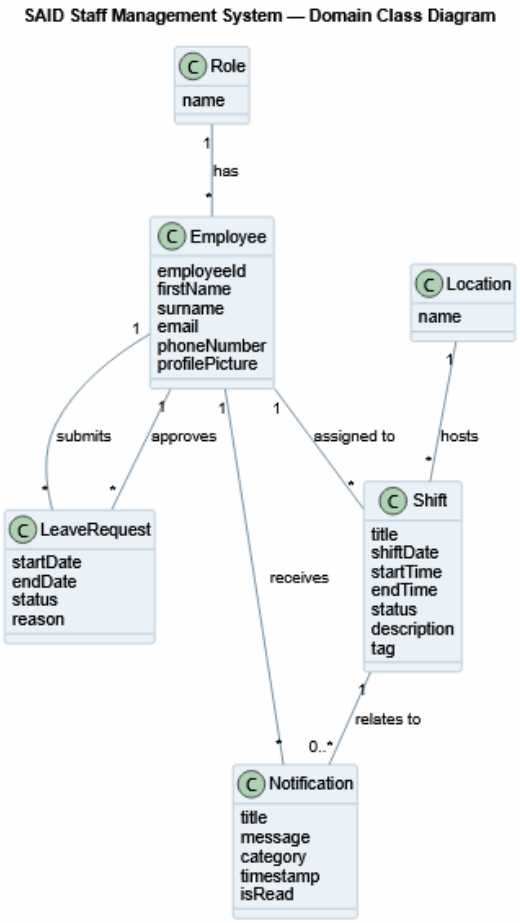
The functional requirements are the basic requirements needed to meet this objective. They include secure authentication, role-based access, staff information management, operational-area management, recurring task templates, daily and weekly roster management, staff allocation and reassignment, roster publication, personal staff schedules, task-status management, notifications, conflict detection, uncovered-duty detection, supervisor dashboards, operational notes, historical records and roster archiving.

The non-functional requirements define the quality criteria needed to ensure that these capabilities function properly. The security, privacy, usability, mobile accessibility, reliability, performance, maintainability, scalability, compatibility, data integrity, backup and recovery, logging, and notification reliability aspects are especially significant due to the fact that the system will be used for real operations in SAID.

Finally, the goal of SAID Operations is not just to digitize an existing roster. The objective is to enhance the operational information creation, communication, action, monitoring and recording.

9.1.9 Analysis artifacts

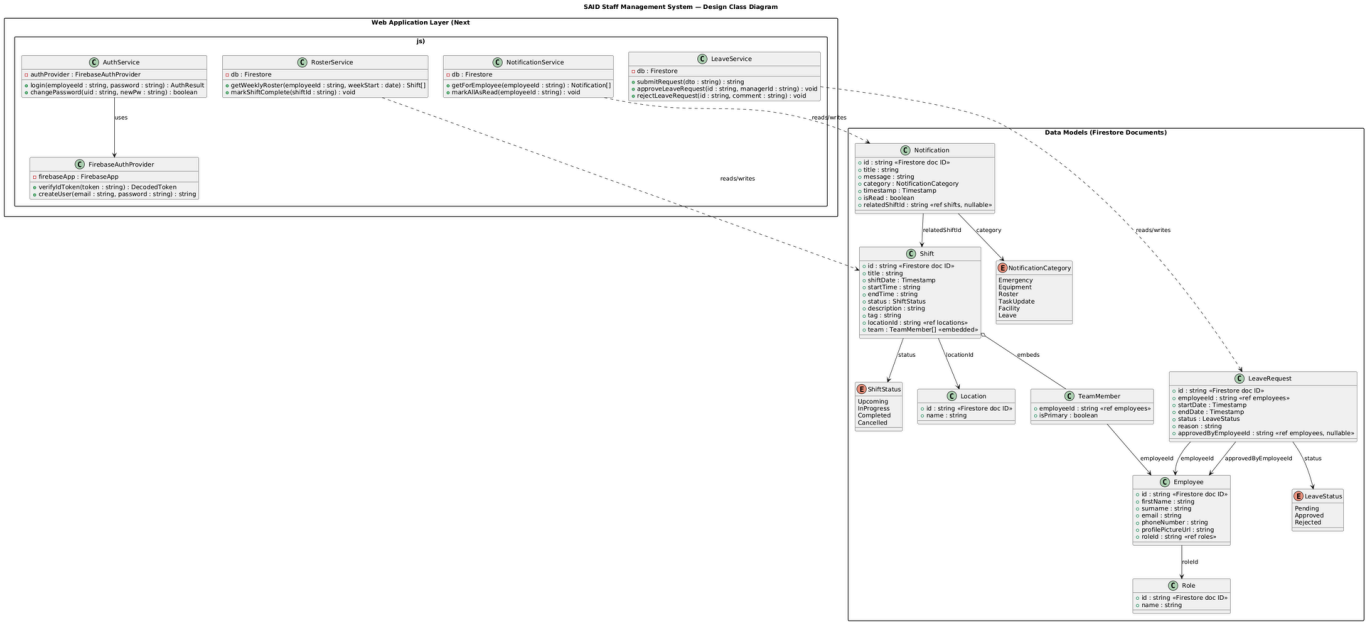
Domain Modelling



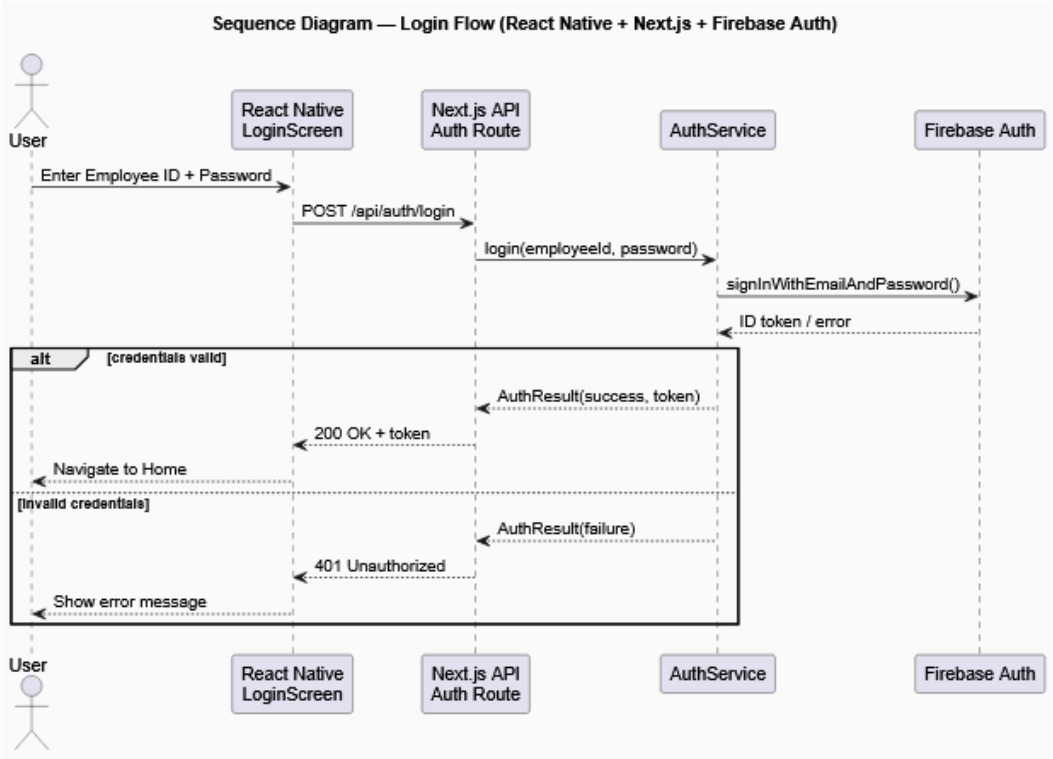
Class	Purpose
Employee	Represents a staff member in the SAID organisation. It stores identifying and contact information such as employee ID, name, email address, phone number and profile picture.
Role	Represents the position that an employee has been given. It enables the system to determine an employee's role or duty inside the company.
Location	Represents a real place where employees work shifts. Multiple shifts can be hosted at each venue.
Shift	Symbolises a planned work shift. The shift title, date, start and end hours, status, description, and tag are among the details it includes. Employees are allocated a shift, which occurs at a designated location.

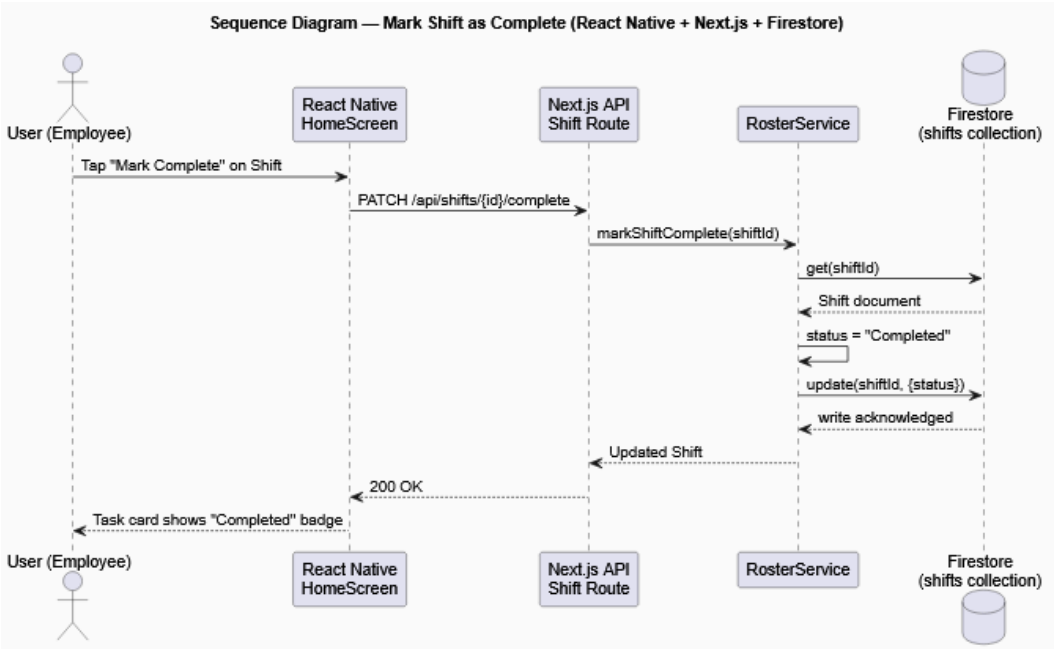
LeaveRequest	Represents a leave request submitted by an employee. It documents the requested dates, current status, justification, and, if relevant, the manager or employee who gave their approval.
Notification	Represents a notification or message given to a worker. Shifts, leave, emergencies, equipment, roster changes, and other staff-management events may all be the subject of notifications.

Design Artifacts

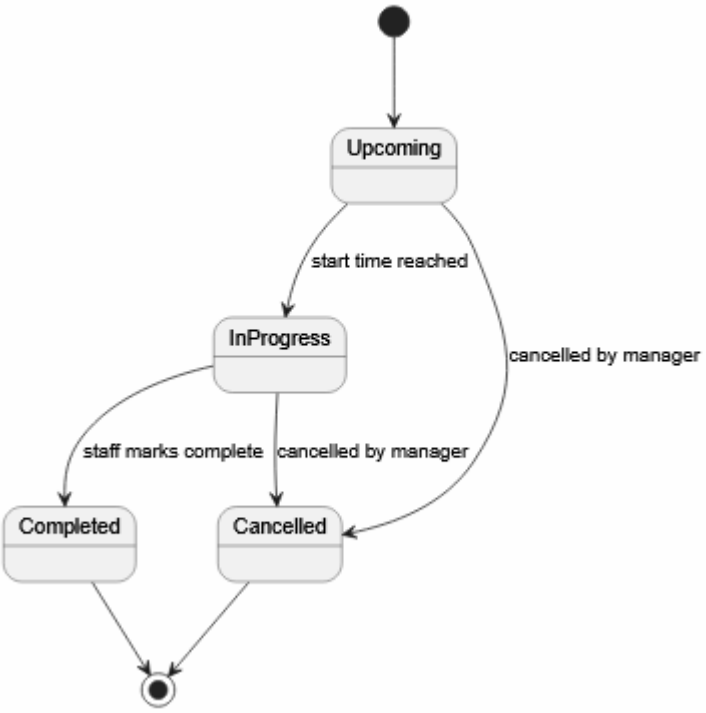


9.1.10 Implementation

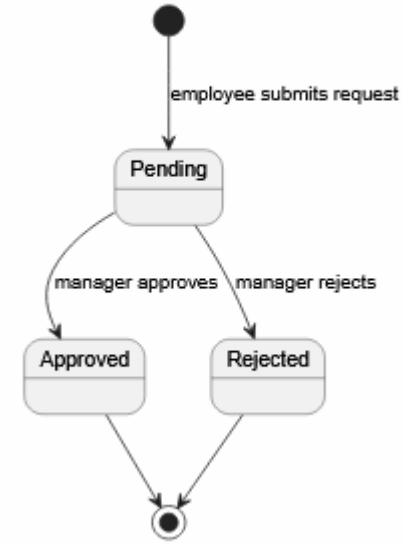


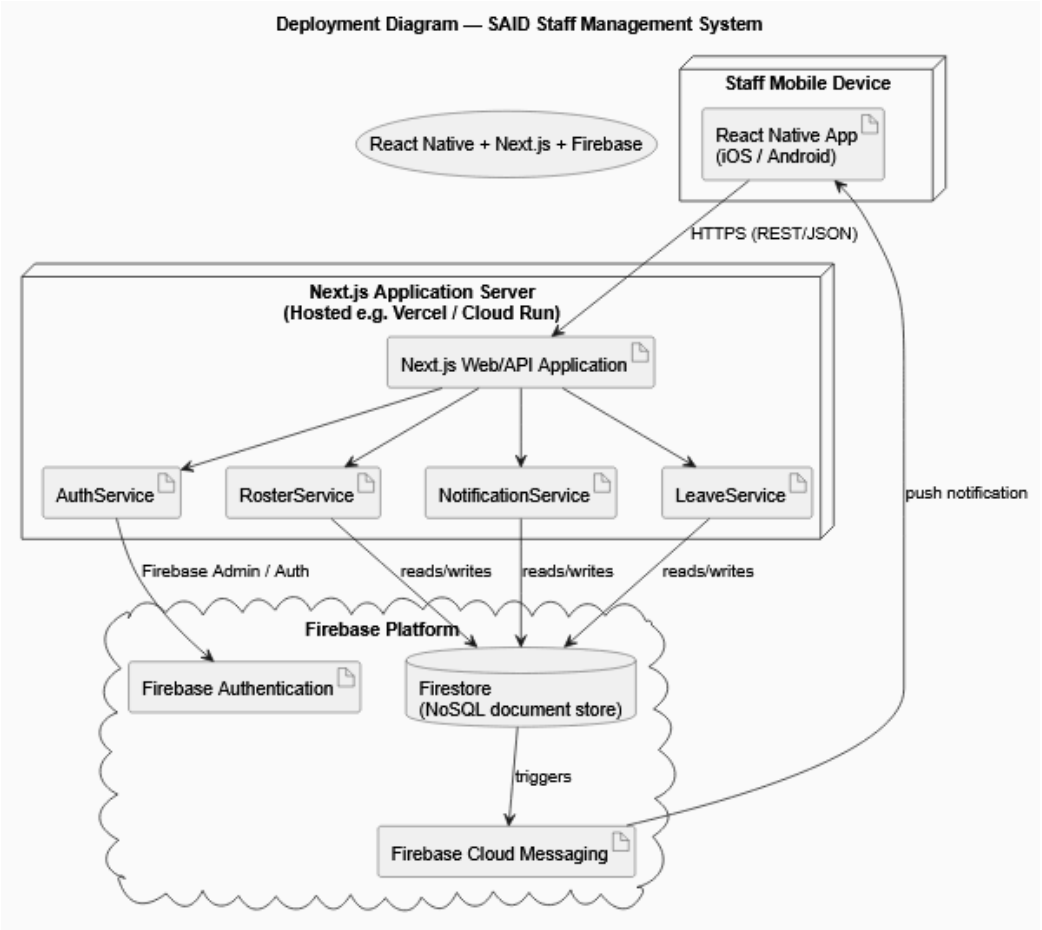


State Diagram — Shift Lifecycle

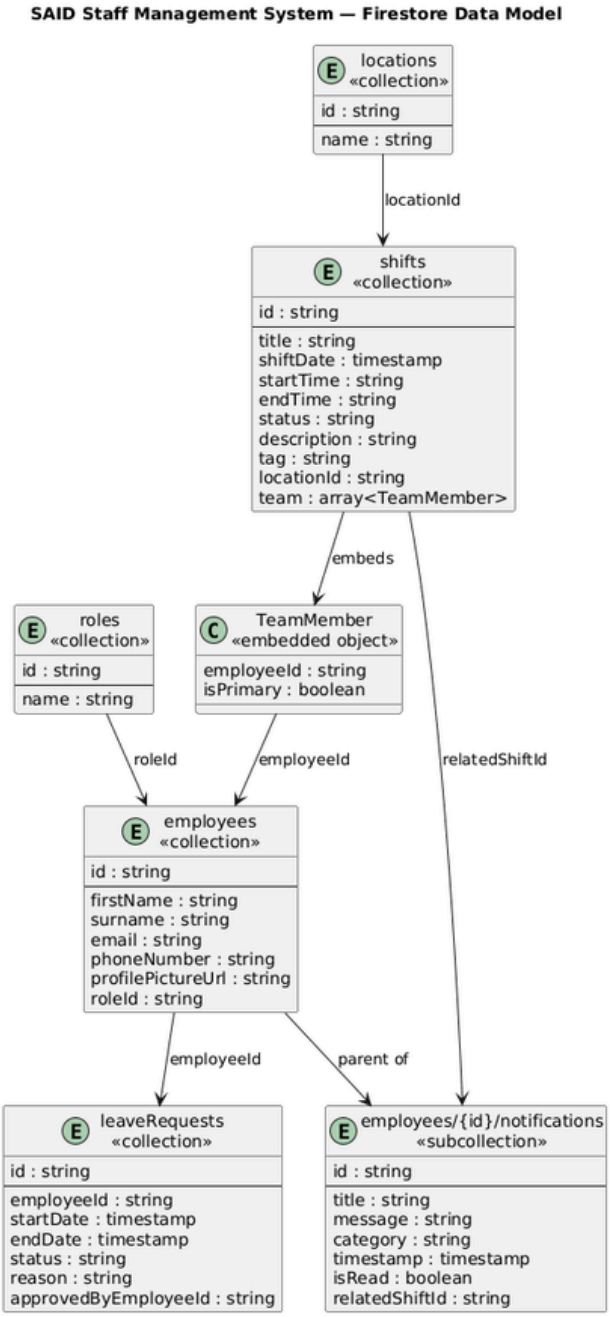


State Diagram — Leave Request Lifecycle




Deployment Diagram — SAID Staff Management System
Staff Mobile Device
React Native + Next.js + Firebase
React Native App
(iOS / Android)
HTTPS (REST/JSON)
Next.js Application Server
(Hosted e.g. Vercel / Cloud Run)
Next.js Web/API Application
AuthService
RosterService
NotificationService
LeaveService
push notification
Firebase Admin / Auth
reads/writes
reads/writes
reads/writes
Firebase Platform
Firebase Authentication
Firestore
(NoSQL document store)
triggers
Firebase Cloud Messaging

9.1.11 Data Schema



Data Schema - Description and Explanation

Firebase Firestore's data organization is depicted in the Firestore Data Model diagram. Since Firestore is a NoSQL document database, collections, documents, and subcollections are used in place of conventional relational tables.

Employees collection

The employees collection stores the main information about each employee. Every employee document contains fields such as:

- id
- firstName
- surname
- email
- phoneNumber
- profilePictureUrl
- roleId

The roleId field links the employee to a document in the roles collection. This avoids storing the complete role information repeatedly for every employee.

Roles collection

The many roles that are accessible throughout the organization are stored in the roles collection. Every role has a name and an ID. Workers use roleId to refer to their roles.

This establishes a relationship that is comparable to a foreign key in a relational database, except Firestore uses a stored ID or document reference instead of a conventional foreign-key constraint.

Shifts collection

The shifts collection stores all scheduled staff shifts. A shift contains information including:

- shift title
- shift date
- start and end times
- status
- description
- tag
- location ID
- assigned team members

The status field represents the current stage of a shift, such as Upcoming, In Progress, Completed or Cancelled.

Locations collection

The physical workspaces that the organization uses are included in the locations collection. Every location has a name and an ID.

The locationId stored in a shift identifies where that shift will take place.

TeamMember embedded object

TeamMember is not a separate collection. Instead, it is an embedded object inside a shift document.

It contains:

- employeeId
- isPrimary

This enables the system to keep track of the workers assigned to a specific shift right within the shift record. The person is identified by their employeeId, and whether they are the main team member for that shift is indicated by their isPrimary.
This approach is appropriate because the team members are closely associated with the shift itself.

leaveRequests collection

The leaveRequests collection stores employee leave applications. Each document contains:

- employeeId
- startDate
- endDate
- status
- reason
- approvedByEmployeeId

The employee who made the request is identified by the employeeId. Whether a request is pending, approved, or rejected is noted in the status. The manager or authorised staff member who handled the request can be identified using approvedByEmployeeId.

Overall schema structure

The schema is designed around the main operations of the SAID Staff Management System: Roles, shifts, locations, leave requests, and notifications are all given to employees.

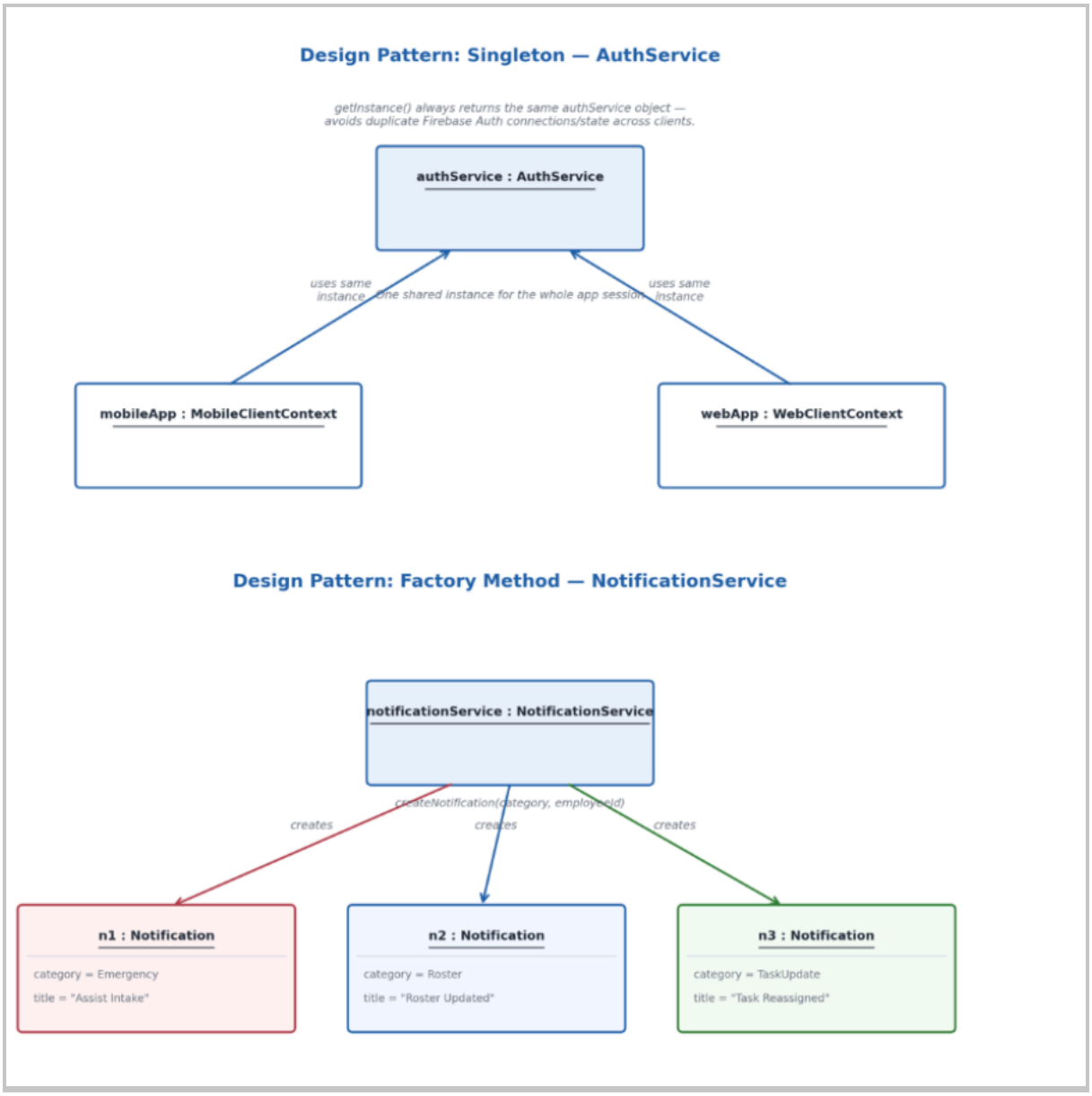
A combination of references and embedded data are used in the design. While TeamMember is directly integrated within a shift since it represents data that is closely relevant to that shift, references like roleId, employeeId, and locationId link related Firestore documents.

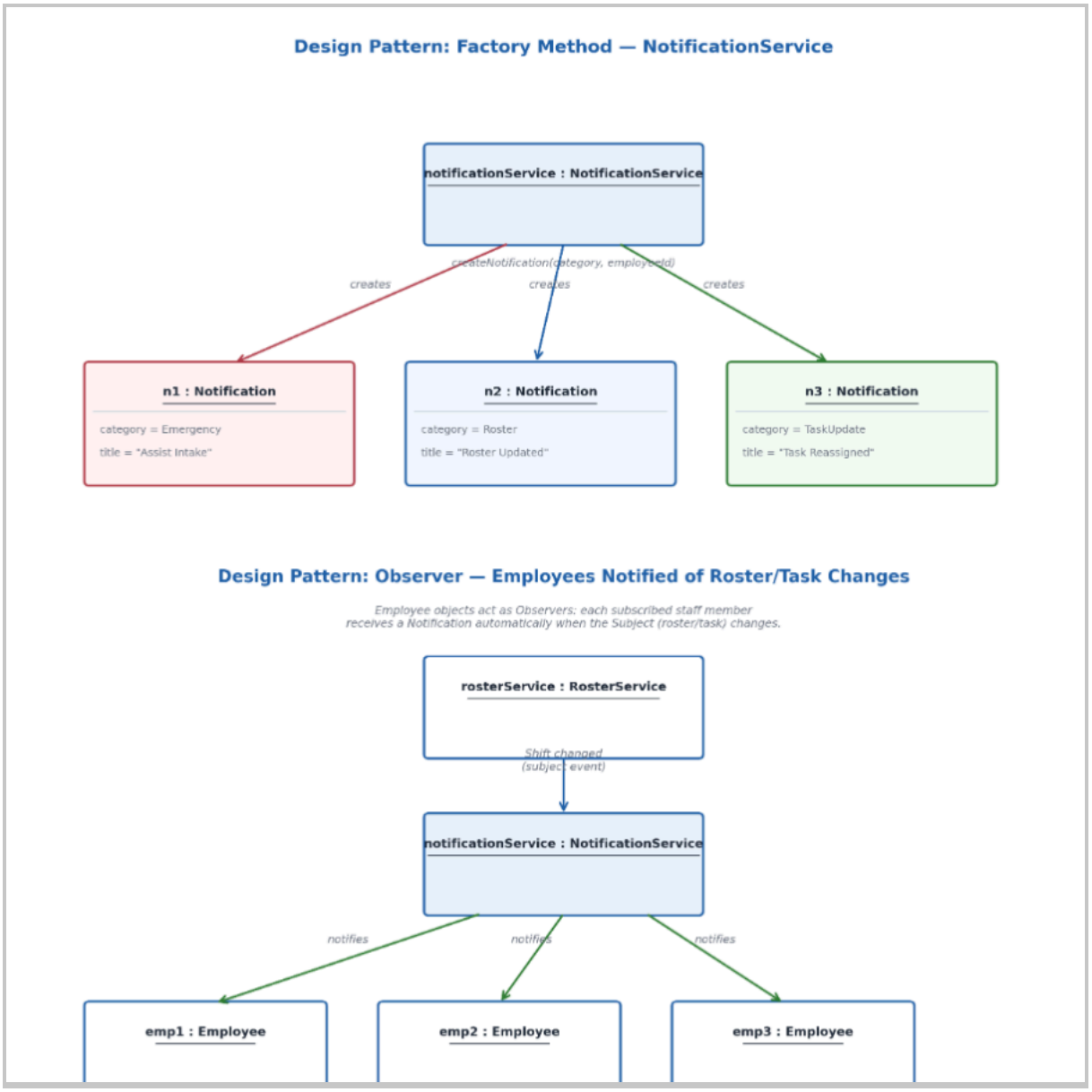
9.1.12 Architecture Artifacts

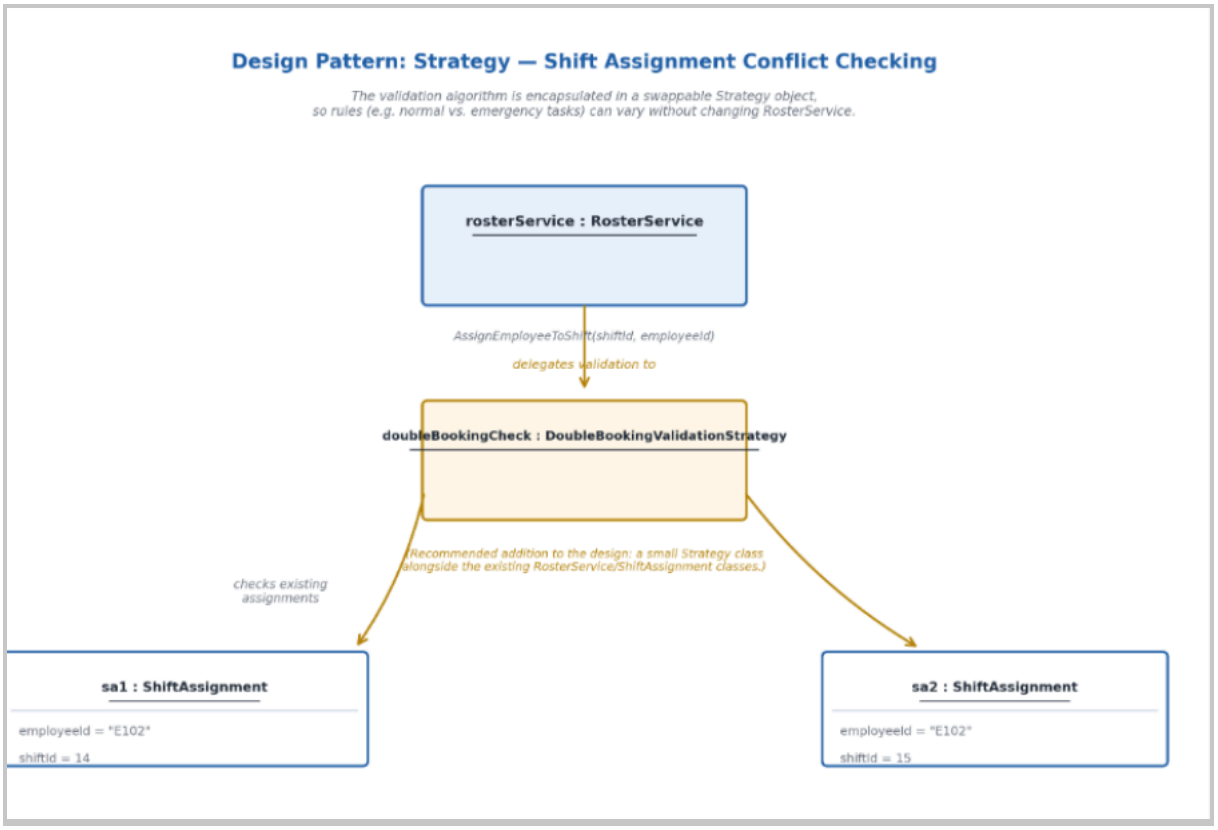
9.3.12 Architecture Artifacts

Design Patterns

Patterns	The use of patterns	Reason for selection
Singleton	Firebase	Becaause
Observer	Real-time updates, especially for task assignments and roster changes	
Factory	To create notification objects	Standardises how different notifications are created, ensuring that each alert/notification carries relevant information no matter how or where in the code it is triggered
Repository	Data Access Layer	
Strategy	Checking if a staff member has a task that is already assigned when assigning them a task	


Design Pattern: Singleton — AuthService
getInstance() always returns the same authService object —
avoids duplicate Firebase Auth connections/state across clients.
authService : AuthService
uses same instance
one shared instance for the whole app session
uses same instance
mobileApp : MobileClientContext
webApp : WebClientContext
Design Pattern: Factory Method — NotificationService
notificationService : NotificationService
createNotification(category, employeeId)
creates
creates
creates
n1 : Notification
category = Emergency
title = "Assist Intake"
n2 : Notification
category = Roster
title = "Roster Updated"
n3 : Notification
category = TaskUpdate
title = "Task Reassigned"


Design Pattern: Factory Method — NotificationService
notificationService : NotificationService
createNotification(category, employeeId)
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n1 : Notification
category = Emergency
title = "Assist Intake"
n2 : Notification
category = Roster
title = "Roster Updated"
n3 : Notification
category = TaskUpdate
title = "Task Reassigned"
Design Pattern: Observer — Employees Notified of Roster/Task Changes
Employee objects act as Observers; each subscribed staff member
receives a Notification automatically when the Subject (roster/task) changes.
rosterService : RosterService
Shift changed
(subject event)
notificationService : NotificationService
notifies
notifies
notifies
emp1 : Employee
emp2 : Employee
emp3 : Employee

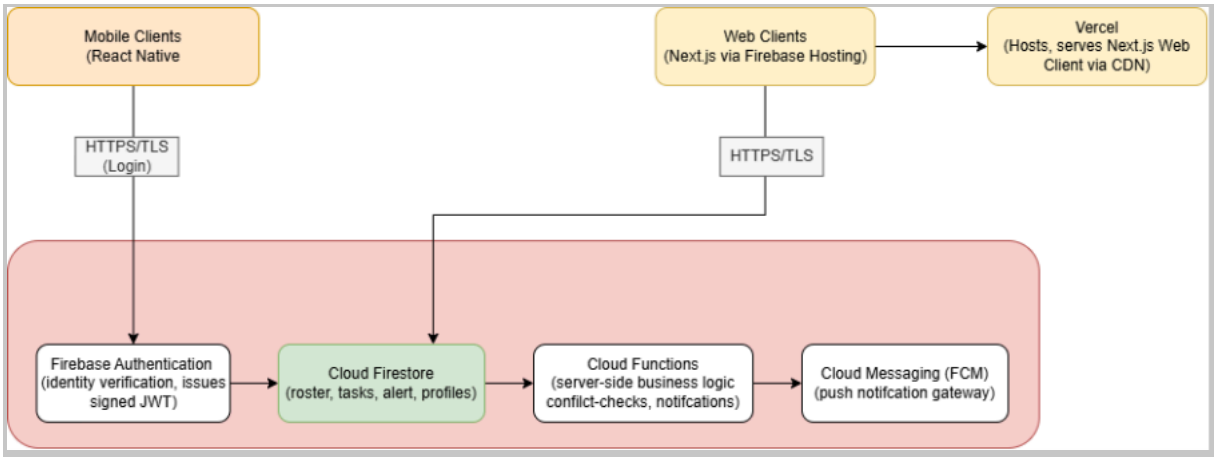


Architecture Pattern

Mobiles Application: used by staff members and supervisors to view tasks, roster, and notifications.

Web Application: used by supervisor and staff members view tasks, roster, and notifications, but supervisors will also use it to view the roster and updated it.

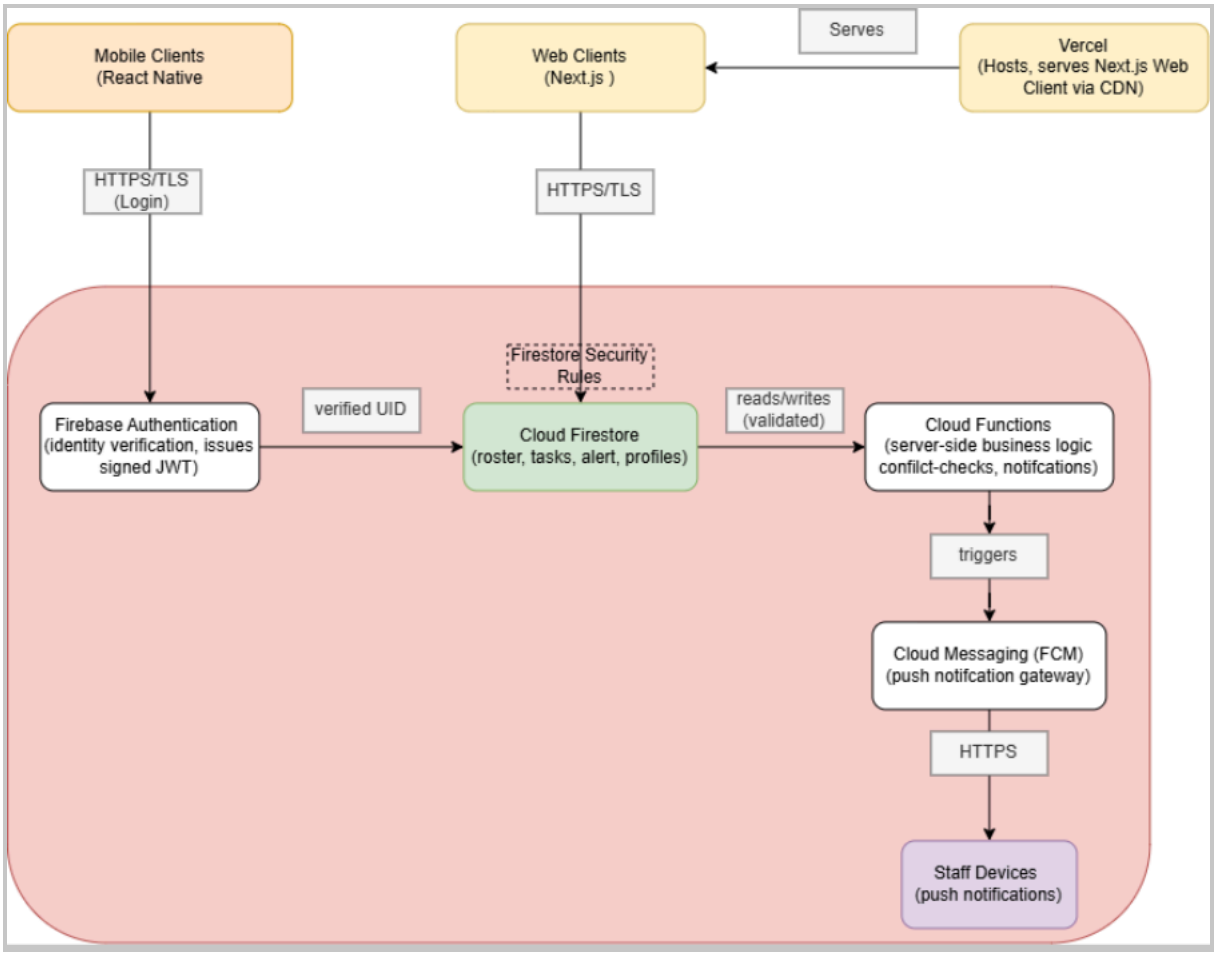
Backend Services (Firebase): a backend used by both platforms' web application and mobile application



Cloud

The application is built on Google Firebase, this is because of how well it works with The Next.js/React stack which will be used to build the web application (Firebase, 2019). These are the Firebase service that will be used in this application.

Service	Purpose
Firebase Authentication	Manages staff registration, login and verification using EmployeeID
Cloud Functions	Server-side business logic and validation, and trigger push notifications
Cloud Firestore	No-SQL real-time database, still will store all the entities for the application. It will also be used.
Firebase Cloud Messaging (FCM)	Delivers push notifications to staff devices
Vercel	Hosts and serves Next.js web application



9.1.13 Security

Security is essential because a data breach could not only affect the organization, but it could be a threat to other parties that are involved with them such as their donors, volunteers, the patients, and the caregivers of the patients. We have tried to minimize the possibility of that happening by applying the standard security principles such as, input validation, least privilege, defence in-depth, and secure-by-default-configuration.

Authentication and Authorization

- Firebase Authentication handles all the staff members login details, and it hashes and stores the credentials, plain text passwords are not stored by the application (Google, 2019).
- Role-based access control (RBAC) is established using Firebase in order to distinguish between staff and supervisor.
- Server-side authorisation is enforced by Firestore Security Rules, this is done on every request (Firebase security rules, n.d.). This ensures that staff members cannot access other staff members information or do things outside the scope of their role.
- Cloud Functions validate a caller's role instead of just accepting role information supplied by the client.

Data Protection

- All the data that is in transit is encrypted using HTTPS/TLS, this is done doe by Firebase itself throughout all its services.
- Firestore encrypts all the data that is at rest.
- Sensitive data fields are only made visible to the specific staff members or the supervisors via Security rules field level restrictions.

Cross-Site Scripting (XSS)

- Next,js is being used to build the web application which by itself does not render dynamic content in JSX, this prevents execution of injected HTML/JavaScript (OWASP, 2021).
- The application prevents DOM manipulation of any user generated content. If rich text is unavoidable, the output will be sanitised before being rendered (OWASP, 2021).
- Every user input is validated and length-limited on both the client-side and server-side (OWASP, 2021).

Risk	Likelihood	Impact	Mitigation

Cross-Site Scripting	Medium	High	Next.js automatically escapes rendered content. Rich text is sanitised before rendering
Injection through malformed input	Low	High	Cloud Functions check and validate input before processing. Firebase SDK uses parametrised queries
Broken authentication	Medium	High	Firebase has and enforces a minimum password policy and secures and hashes credentials.
Sensitive data exposure	Medium	Medium	Sensitive data is never logged on the client's side, and the Firebase Security rules restrict field collection to the role of the user.
Security misconfiguration	Medium	High	Firestore is deployed is strict rules, it is never left open in ‘test’ mode
Excessive automated requests	Low	Medium	Cloud Functions apply rate limiting. Firebase verifies the origin of a request to make sure it comes from a real application.
Insecure direct object reference	Low	High	All reads and writes are checked for the users UID to this prevents users accessing other users' information.

Man-in-the-middle attack	Low	High	All traffic is forced through HTTPS/TLS, this is through the entire Firebase ecosystem. HTTP endpoints are not accepted

9.1.14 DevOps

Devops is the approach of combining automated testing, continuous integration, controlled delivery and version control in order to produce high-quality software quickly (IBM, 2021). Rapid and reliable development is essential when building software, in our case the organization will need this because the entire organization will be dependent on this software in order to manage the staff members and the work that is done. This is the reason why our team will be using GitHub for version control and we will use GitHub actions for continuous integration and continuous delivery (CI/CD) to maintain the quality of the application, and for speedy delivery.

Branching Model and Workflow

Branch	Purpose	Control
main	Production ready application version	Protected branch, pull requests equired, reviews and automated test required
development	Integration and Staging	Continuous Integration (CI) is required, merge through reviewed pull requests
feature	Individual feature	Short-lived, linked to a task, pull request into development

The model that we went with is a Gitflow model. The Gitflow model allows for the application in the main branch to be deployed while the team may be working on new features or bug fixes, this does not disrupt the current deployed version and it allows for changes to only be released once they are reviewed and tested (Github.io, 2018).

GitHub Actions Pipeline

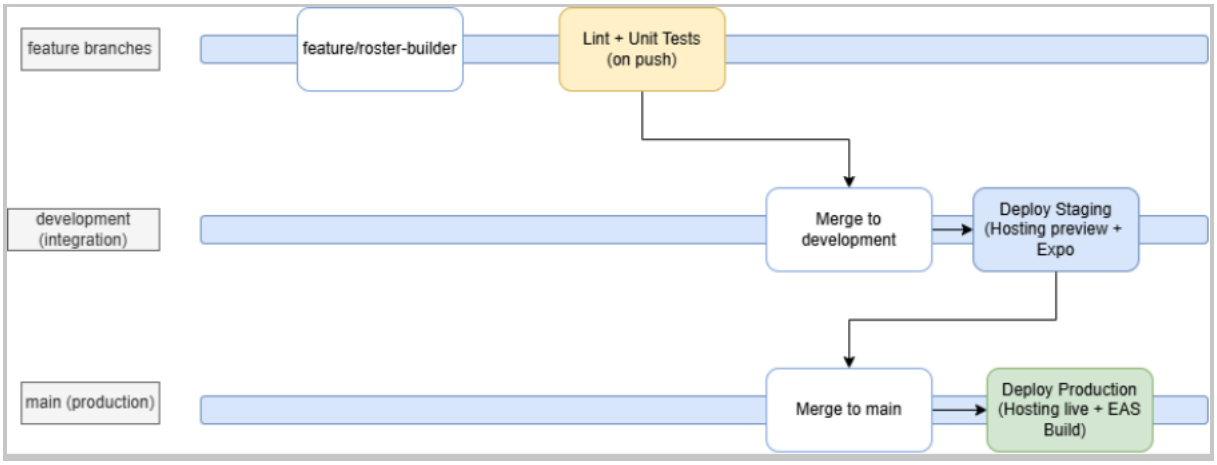
The pipeline is triggered automatically when there are push and pull requests, and it does the following:

Lint & Static: ESLint/Typescript checks are done on every push into a feature branch and run when there is a pull request. This will catch any style or type errors early.

Automated Testing: Jest unit tests are run automatically on every pull request into the development branch. Cloud functions are tested against the Firebase Emulator Suite; this is done to check the business logic without touching the production data.

Merge and Staging: Once a pull request has passed all the checks and is approved then it is merged into the develop branch. This automatically triggers a deployment into the Firebase Hosting preview channel, and the internal Expo build for the mobile application; this allows the team to test the integrated application before deployment.

Production Release: A pull request from the development to the main branch triggers all the tests to get re-run and if they pass, the web application is deployed to the live Firebase Hosting channel and the mobile application is built and submitted via the EAS Build by Expo.



9.1.15 Running Costs

SAID Ops is planned as a cloud-based system with a low starting cost and enough flexibility to grow if usage increases. The estimate below covers the expected monthly cost of hosting, database services, authentication and notifications. Development time and team fees are excluded because they are not ongoing costs of running the system.

The planned technology stack is:

- React Native for the staff mobile application.
- Next.js for the supervisor web application.
- Firebase for the database, authentication and notifications.
- Vercel for hosting the supervisor web application.

Costs are presented in South African Rand (ZAR). Firebase and Vercel publish most of their pricing in United States Dollars (USD), so the estimate uses the South African Reserve Bank rate of US$1 = R16.1744 recorded on 14 August 2026 (South African Reserve Bank, 2026). The exchange rate and supplier prices should be checked again before the final project submission and before production deployment.

9.1.15.1 Costing Assumptions

The estimates are based on the following assumptions:

- One Vercel Pro developer seat will be used to host and manage the Next.js supervisor web application. Additional developer seats are not included.
- Cloud Firestore will be the main database and will use the Johannesburg region where appropriate.
- Firebase Authentication will use email and password login rather than paid SMS authentication.
- Firebase Cloud Messaging will be used for push notifications and operational alerts.
- Production notification triggers may require Firebase Cloud Functions or another trusted server environment. No separate monthly charge has been included because the final implementation choice has not yet been confirmed; this must be reviewed during implementation.
- SAID will use an existing domain or subdomain, so no new domain registration cost has been included.
- The system will mainly store small, text-based operational records such as staff details, rosters, tasks, assignments and notifications.
- Large media files and veterinary medical records will remain outside SAID Ops.
- The calculations use approximately 22 working days per month.
- Each active user is estimated to generate about 20 Firestore document reads and five document writes per working day.
- For the 10,000-user scenario, the estimate assumes about 2 GiB of stored operational data. One GiB is covered by the free allowance and one GiB is treated as billable storage.

Cloud Firestore currently includes a free allowance of 50,000 document reads per day, 20,000 writes per day, 20,000 deletes per day, 1 GiB of stored data and 10 GiB of outbound data transfer per month (Google Cloud, 2026). Firebase Cloud Messaging has no standard message-delivery charge, while the authentication services expected for SAID Ops should remain within the available free allowance at the user levels considered here (Firebase, 2026).

Vercel Pro is priced at US$20 per month. At the exchange rate used in this estimate, this is approximately R323.49 per month. The plan also includes US$20 of monthly infrastructure usage credit (Vercel, 2026).

9.1.15.2 Estimated Initial Monthly Cost

At SAID's expected initial scale, Firebase usage should remain within the available free allowances. The estimated starting cost is therefore:

Service	Cost basis	Estimated monthly cost
Vercel Pro	One developer seat at US$20 per month	R323.49
Cloud Firestore	Expected initial usage remains within the free allowance	R0.00

Firebase Authentication	Email/password usage remains within the expected free allowance	R0.00
Firebase Cloud Messaging	No standard message-delivery charge	R0.00
Domain	Existing SAID domain or subdomain	R0.00
Estimated initial total	Monthly operating estimate	R323.49

React Native and Next.js do not add separate monthly infrastructure charges. Their operating costs are linked to the hosting and cloud services used by the completed applications.

9.1.15.3 Cost at Different User Scales

The module requires the solution to be cost at different levels of usage. The table therefore compares 100, 1,000 and 10,000 active users. The larger scenarios do not represent SAID's expected employee numbers; they show how the selected technology could scale if the system were used more widely in future.

Active users	Reads per working day	Writes per working day	Estimated Firebase cost	Vercel Pro	Estimated monthly total
100	2,000	500	R0.00	R323.49	R323.49
1,000	20,000	5,000	R0.00	R323.49	R323.49
10,000	200,000	50,000	R28.05	R323.49	R351.54

At 100 and 1,000 users, the estimated reads and writes remain below the current daily Firestore allowances. At 10,000 users, the system would exceed those allowances. Using 22 working days, the estimated billable usage would be 3.3 million reads and 660,000 writes per month.

Based on the current Firestore Standard Edition prices and the exchange rate used above, the 10,000-user estimate consists of approximately R16.01 for reads, R9.61 for writes and R2.43 for one billable GiB of storage. This produces an estimated Firebase cost of R28.05 and a total monthly cost of approximately R351.54 (Google Cloud, 2026).

These figures assume that SAID Ops continues to store mainly small operational records. Large file storage, heavier network traffic, additional cloud services or changes to supplier pricing would increase the final cost.

9.1.15.4 Two-Year Cost Projection

The client discussions confirmed that SAID Ops is intended as an internal staff operations system, but SAID has not yet supplied a formal two-year user-growth forecast. The figures below are therefore Task 1 planning assumptions and must be confirmed with the client before final submission.

Period	Planning assumption	Estimated monthly cost	Estimated annual cost
Year 1	Up to 100 active users	R323.49	R3,881.88
Year 2	Up to 250 active users	R323.49	R3,881.88
Two-year total	Initial planning estimate	-	R7,763.76

The monthly estimate stays unchanged between Year 1 and Year 2 because both assumed user levels remain within the current Firebase free allowances. The two-year amount is not a fixed quotation. It should be recalculated if SAID confirms different growth figures, the system design changes, supplier prices change or the USD/ZAR exchange rate moves significantly.

9.1.15.5 Cost Monitoring and Control

Cloud usage should still be monitored even though the expected monthly cost is low. The following controls are recommended:

- Review Firestore reads, writes, storage and network usage regularly.
- Configure billing alerts if the Firebase Blaze pay-as-you-go plan is enabled.
- Monitor Vercel usage for unexpected traffic or infrastructure increases.
- Use efficient Firestore queries and avoid unnecessary repeated reads.
- Store only the operational information required by SAID Ops.
- Avoid unnecessary large file uploads.
- Use Firebase Cloud Messaging for operational notifications instead of paid SMS services.
- Review costs whenever major functionality or a new cloud service is introduced.
- Update the ZAR estimate when exchange rates or supplier prices change materially.

Keeping the operating cost manageable is important because SAID is a non-profit organisation. The selected technology stack offers a low starting cost while still allowing the system to grow if the organisation's needs increase.

9.1.16 Change Management

Introducing SAID Ops will change how staff rosters, task allocations and operational updates are managed at The Society for Animals in Distress (SAID). A structured change process is needed so that the system can be introduced without disrupting day-to-day operations.

SAID Ops is not intended to replace every system currently used by the organisation. Clinical systems such as ezyVet will continue to manage veterinary and patient information. SAID Ops will focus on staff scheduling, task allocation, roster visibility and operational communication. This keeps the change manageable and reduces the amount of new work introduced at one time.

9.1.16.1 Organisational Adoption

SAID should adopt the system because it responds directly to problems identified during the client discussion and the review of the existing roster. Supervisors currently coordinate duties across admissions, wards, theatre preparation, stock checks, kennels, outreach and other recurring operational areas.

SAID Ops will provide one place where supervisors can create and update rosters, assign staff, identify unassigned duties and communicate changes. This should make scheduling easier to manage and give supervisors a clearer view of daily work.

Management involvement will be important throughout development and implementation. Supervisors and other relevant SAID representatives should review the prototypes, test the roster functions and confirm that the system reflects real working practices.

Involving SAID throughout the process will reduce the risk of delivering a system that does not suit the organisation. Kotter's approach emphasises the need for a clear reason for change, involvement of the people affected and the removal of barriers to adoption (Kotter, n.d.). These principles will guide client reviews, testing, feedback and rollout.

9.1.16.2 User Adoption

The main users will be administrators, supervisors and staff members. Adoption will depend on whether they find the system useful, easy to understand and practical in their day-to-day work.

Davis (1989) identifies perceived usefulness and perceived ease of use as important influences on the acceptance of information systems. SAID Ops will therefore simplify existing activities instead of creating unnecessary work.

Staff will be able to view their own daily duties, weekly roster and important alerts without searching through the full organisation-wide roster. Supervisors will have additional functions for managing assignments, changing rosters and monitoring operational responsibilities.

The interface will use clear navigation, familiar SAID terminology and straightforward task statuses such as Upcoming, In Progress, Paused and Completed. Early user testing will allow staff to become familiar with the system and identify any instructions or screens that need improvement.

9.1.16.3 Adoption Strategy

A phased rollout is proposed instead of immediately replacing the current roster process:

1. Present the prototype to SAID and confirm the main functions, design and workflows.
2. Refine the system using agreed client feedback.
3. Allow the supervisors who manage the roster to test scheduling, recurring duties and staff allocation.
4. Allow a small group of staff members to test the mobile application and report usability problems.
5. Run the current roster and SAID Ops in parallel for a limited transition period.

6. Make SAID Ops the primary platform once the main functions have been tested, training has been completed and SAID approves the transition.

9.1.16.4 Transition from the Current Process

Only the operational information needed by SAID Ops will be captured. Veterinary records and existing patient information will remain in SAID's clinical systems and will not be migrated. Initial setup information may include:

- Staff profiles and user roles.
- Operational areas or departments.
- Common and recurring duties.
- Staff allocation information.
- Current roster information.
- User accounts and access permissions.

Previous roster spreadsheets should be retained as historical records rather than transferring every old roster into the new database. This limits unnecessary migration work and reduces the risk of importing outdated information. Before rollout, supervisors should check the staff details, operational areas, current roster and recurring task templates.

9.1.16.5 Training Plan

Training will be based on each group's responsibilities so that users focus on the functions they actually need. The durations below are proposed and should be confirmed with SAID when the rollout date and staff availability are known.

User group	Training focus	Proposed method and duration	Completion evidence
Administrators	User accounts, staff information, permissions and basic configuration	60-90 minute guided practical session led by the project team	Complete the initial account and configuration checklist
Supervisors	Rosters, duty allocation, alerts, roster changes and scheduling issues	60 minute demonstration followed by supervised practice	Create a sample roster and complete a task-allocation exercise
Staff members	Login, daily and weekly duties, alerts and task-status updates	30-45 minute practical demonstration supported by the user guide	Complete the guided mobile workflow

Training examples should reflect familiar SAID work, such as admissions, ward duties, theatre preparation, stock checks and outreach. A short user guide with screenshots and step-by-step instructions should be available after training. Where possible, training should take place close to rollout so that users can apply what they have learned.

9.1.16.6 Handover and Production Transition

The system should only be handed over after the main functions have been tested and the client agrees that it is ready. Before handover, the project team should:

- Complete functional and usability testing and correct important defects.

- Confirm that the production system is hosted, accessible and connected to the required database and authentication services.
- Configure the required administrator and supervisor accounts.
- Provide the final user guide, system documentation and a list of any known unresolved issues.
- Transfer the source repository and relevant system ownership information to the agreed SAID representative.
- Transfer hosting and administrator access through SAID-owned accounts where possible.
- Transfer configuration information securely without placing passwords or private keys in the written report.
- Conduct the agreed training sessions and demonstrate the final system.
- Confirm the temporary fallback process to be used if the production system is unavailable.

The handover must include both the working system and enough supporting information for SAID to continue using it after the WIL project ends. The organisation should not depend on the student team for normal administration and day-to-day use.

9.1.16.7 Support Strategy

A proposed two-week support period should follow go-live, although the final period must be agreed with SAID. This gives the team time to resolve problems that only become visible during normal use. Users should be encouraged to report:

- Login or account-access problems.
- Incorrect roster information.
- Missing or incorrect task assignments.
- Problems updating task statuses.
- Notification or alert problems.
- Usability difficulties.
- Unexpected system errors.

Problems that prevent staff from accessing their roster or prevent supervisors from managing essential assignments should be treated as high priority. Minor interface suggestions can be recorded for later improvement. At the end of the agreed support period, normal administration should move to the designated SAID administrator or supervisor.

9.1.16.8 Change Risks and Mitigation

Risk	Possible impact	Mitigation
Staff are hesitant to use the new system	Users may continue relying on the previous roster process	Involve staff in testing, explain the purpose of the system and provide practical training
Users have different levels of technical experience	Some users may struggle with basic functions	Keep the interface simple and provide role-specific training and guidance
Incorrect information is captured during setup	Staff may receive incorrect tasks or roster information	Require supervisors to verify staff, task and roster information before rollout

Both processes continue indefinitely	Different roster versions may become inconsistent	Use parallel operation for a limited period and confirm when SAID Ops becomes the primary platform
Internet or service interruptions occur	Staff may temporarily lose access to roster information	Agree on a temporary fallback process for essential duties
Important functionality is missing	The system may not fully support SAID's workflow	Conduct client reviews and user testing before full rollout
Key users are unavailable for training	Some staff may not understand the required functions	Provide a user guide and arrange an additional session where needed

9.1.16.9 Proposed Adoption Timeline

The timeline is an initial plan and may change according to client feedback, development progress and the availability of SAID staff.

Phase	Main activities	Proposed duration
1. Client validation	Present the prototype, confirm requirements and obtain client feedback	Week 1
2. Development and refinement	Develop approved functionality and implement agreed changes	Weeks 2-3
3. Supervisor testing	Test roster creation, staff allocation, alerts and administration	Week 4
4. Staff user testing	Allow a small staff group to test the mobile application and provide feedback	Week 4
5. Training and setup	Configure users, prepare the guide and conduct role-based training	Week 5
6. Parallel rollout	Use SAID Ops while retaining the existing roster temporarily as a fallback	Week 5
7. Production adoption	Make SAID Ops the primary platform for roster and task management	Week 6
8. Initial support and review	Resolve early issues, review adoption and complete the support handover	Weeks 6-7

9.1.16.10 Measuring Successful Adoption

Successful adoption is not only about deploying the software. SAID must also be able to use the system effectively. The measures below are proposed indicators and should be confirmed with the client.

Desired outcome	How it can be measured	Suggested evidence

Supervisors can publish rosters	Review whether the weekly roster can be created and published without project-team assistance	Completed roster and supervisor feedback
Staff can access their duties	Check whether pilot users can view daily and weekly assignments	Pilot checklist and user feedback
Task progress is updated	Review whether assigned tasks move through the approved statuses	Task-status records
Changes and alerts are noticed	Confirm that test users can locate roster changes and operational alerts	User-testing results
Support needs reduce	Compare the number and type of support requests during the support period	Issue log
Major rollout issues are resolved	Check that high-priority defects have been closed or have an agreed action	Final handover issue list

Feedback should be collected from supervisors and staff after the pilot and again near the end of the support period. The findings can be used to decide whether further training, interface improvements or future functionality are needed.

9.1.17 Appendices

Declaration of authenticity

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